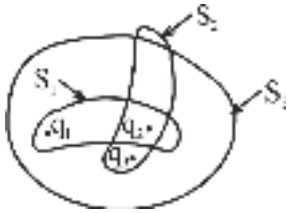
PHYSICS

1) Three charges $q_1 = 1 \times 10^{-6}$ C, $q_2 = 2 \times 10^{-6}$ C and $q_3 = -3 \times 10^{-6}$ C have been placed as shown in figure. Then the net electric flux will be maximum for the surface?



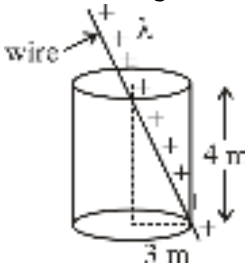
(1) S_1

(2) S_2

(3) S_3

(4) Same for all three

2) Linear charge density of wire is 8.85 µC/m Radius and height of the cylinder are 3m and 4 m. Then find the flux passing through the cylinder :-



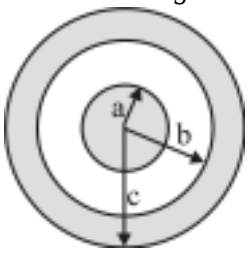
(1) 5×10^6 volt-m

(2) 3×10^6 volt-m

(3) 4×10^6 volt-m

(4) None

3) A solid conducting sphere of radius a has a net positive charge 2Q. A conducting spherical shell of inner radius b and outer radius c is concentric with the solid sphere and has a net charge –Q. The surface charge density on the inner and outer surfaces of the spherical shell will be :-



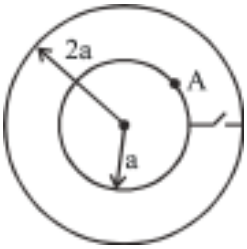
(1) $-\frac{2Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

(2) $-\frac{Q}{4\pi b^2}, \frac{Q}{4\pi c^2}$

(3) $0, \frac{Q}{4\pi c^2}$

(4) None of the above

4) Two metallic concentric shells are shown. Before closing the switch charge on inner shell is Q and charge on outer shell is 2Q. Potential at point 'A' is V. What will be potential at same point after closing the switch :-



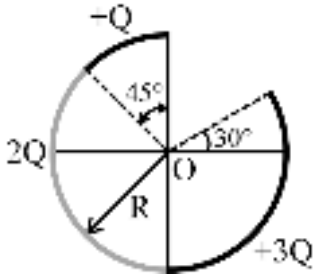
(1) V

(2) 2V

(3) $\frac{3}{4}V$

(4) $\frac{2}{3}V$

5) Figure shows three circular arcs, each of radius R and total charge as indicated. The net electric potential at the centre of curvature is :



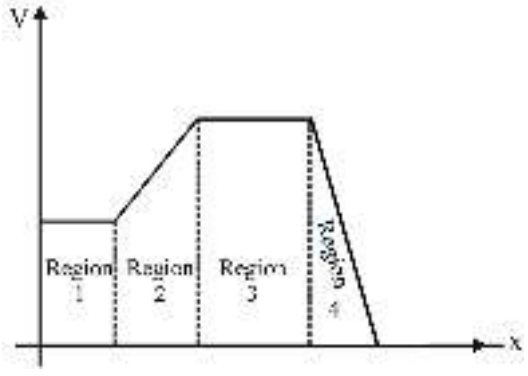
(1) $\frac{Q}{2\pi\varepsilon_0 R}$

(2) $\frac{Q}{4\pi\varepsilon_0 R}$

(3) $\frac{2Q}{\pi\varepsilon_0 R}$

(4) $\frac{Q}{\pi\varepsilon_0 R}$

6) Potential versus distance x graph is shown for a region. Correct order of magnitude of electric field in regions respectively is:-

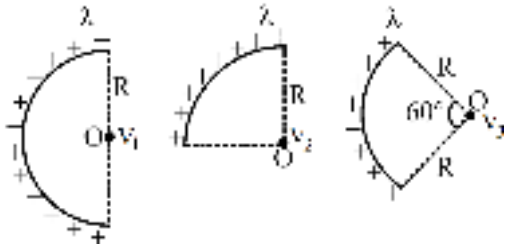


(1) $E_1 > E_2 > E_3 > E_4$

(2) $E_4 > E_3 > E_2 > E_1$

(3) $E_4 > E_2 > E_1 = E_3$

(4) $E_2 > E_4 > E_1 = E_3$

7) Fig. Shows 3 uniformly charged arcs of equal radii. Then correct order of potential at their centres is 

(1) $V_1 = V_2 = V_3$

(2) $V_1 > V_2 > V_3$

(3) $V_1 < V_2 < V_3$

(4) $V_1 = 2V_2 = 2V_3$

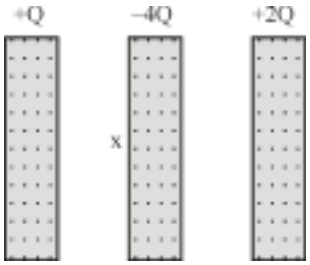
8) 

Figure shows three parallel conducting plates placed close to each other. Then charge on surface x is –

(1) $+\frac{3Q}{2}$

(2) $-\frac{3Q}{2}$

(3) $+\frac{5Q}{2}$

(4) $-\frac{5Q}{2}$

9) Figure shows a family of equipotential lines between which an electron is moved as shown in

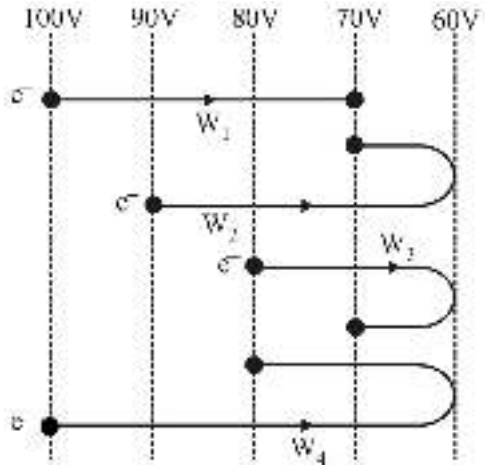


figure. Then correct order of work done in the process is :

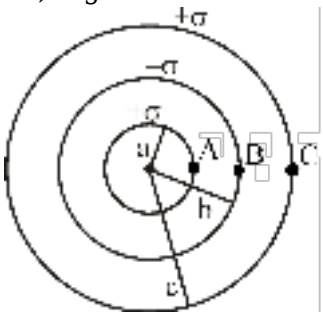
(1) $W_1 > W_2 > W_3 > W_4$

(2) $W_1 < W_2 < W_3 < W_4$

(3) $W_1 > W_2 = W_3 > W_4$

(4) $W_1 > W_2 = W_4 > W_3$

10) Figure shows three thin concentric metallic shells. If potential at surface A & C are equal then



(1) $c = \frac{a+b}{2}$

(2) $c = a + b$

(3) $a + b + c = 0$

(4) $c + 2a + b = 0$

11) Amount of work done in moving a charge q_0 from centre of ring to an axial point at $x = \sqrt{8}a$ will be, if charge given to ring is Q and 'a' being the radius:-

(1) $\frac{Qq_0}{6\pi\varepsilon_0 a}$

(2) $-\frac{Qq_0}{6\pi\varepsilon_0 a}$

(3) Zero

(4) $\frac{Qq_0}{4\pi\varepsilon_0 a}$

12)

Two points A & B are located 2m distance apart in uniform electric field of E = 10 V/m. Then potential difference between points A & B may be :

(1) 25 V

(2) 30 V

(3) 16 V

(4) 28 V

13) An arc of radius r carries charge. The linear density of charge is λ and the arc subtends a angle $\frac{\pi}{3}$ at the centre. What is electric potential at the centre.

(1) $\frac{\lambda}{4\varepsilon_0}$

(2) $\frac{\lambda}{8\varepsilon_0}$

(3) $\frac{\lambda}{12\varepsilon_0}$

(4) $\frac{\lambda}{16\varepsilon_0}$

14) A charged particle q is shot towards another charged particle Q which is fixed, with a speed v. It approaches Q upto a closest distance r and then returns. If q were given a speed 2v, the closest distance of approach would be



(1) r

(2) 2r

(3) r/2

(4) r/4

15) A solid conducting sphere having a charge Q is surrounded by an uncharged concentric conducting hollow spherical shell. Let the potential difference between the surface of the solid sphere and that of the outer surface of the hollow shell be V. If the shell is now given a charge of –2Q, the new potential difference between the same two surfaces is

(1) V

(2) 2V

(3) 4V

(4) – 2V

16) Electric potential is given by $V = 6x - 8xy^2 - 8y + 6yz - 4z^2$
Then electric force acting on 2C point charge placed on origin will be

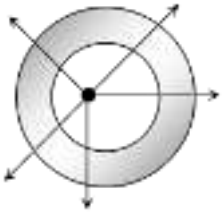
(1) 2N

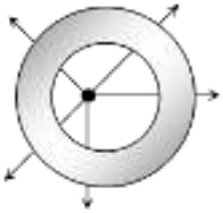
(2) 6N

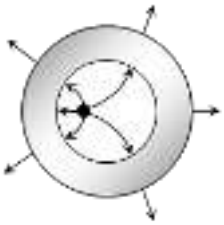
(3) 8N

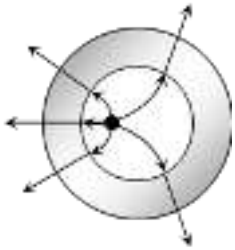
(4) 20N

17) A metallic shell has a point charge ‘q’ kept inside its cavity. Which one of the following diagrams correctly represents the electric lines of forces

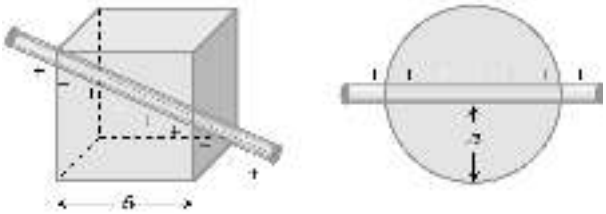
(1) 

(2) 

(3) 

(4) 

18) A linear charge having linear charge density λ, penetrates a cube diagonally and then it penetrate a sphere diametrically as shown. What will be the ratio of flux coming cut of cube and sphere



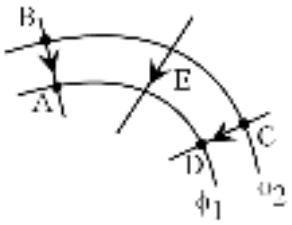
(1) $\frac{1}{2}$

(2) $\frac{2}{\sqrt{3}}$

(3) $\frac{\sqrt{3}}{2}$

(4) $\frac{1}{1}$

19) In moving from A to B along an electric field line, the electric field does 6.4×10^{-19} J of work on an electron. If ϕ_1, ϕ_2 are equipotential surfaces, then the potential difference ($V_C - V_A$) is



(1) – 4V

(2) 4V

(3) Zero

(4) 64 V

20) Two thin wire rings each having a radius R are placed at a distance d apart with their axes coinciding. The charges on the two rings are +q and –q. The potential difference between the centres of the two rings is :

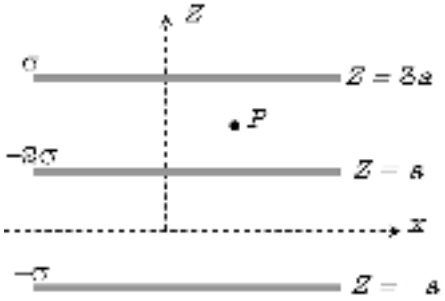
(1) Zero

(2) $\frac{q}{4\pi\varepsilon_0}\left[\frac{1}{R}-\frac{1}{\sqrt{R^2+d^2}}\right]$

(3) $qR/4\pi\varepsilon_0 d^2$

(4) $\frac{q}{2\pi\varepsilon_0}\left[\frac{1}{R}-\frac{1}{\sqrt{R^2+d^2}}\right]$

21) Three infinitely long charge sheets are placed as shown in figure. The electric field at point P is



(1) $\frac{2\sigma}{\varepsilon_0}\hat{k}$

(2) $-\frac{2\sigma}{\varepsilon_0}\hat{k}$

(3) $\frac{4\sigma}{\varepsilon_0}\hat{k}$

(4) $-\frac{4\sigma}{\varepsilon_0}\hat{k}$

22) A charge q is placed at the centre of the open end of cylindrical vessel. The flux of the electric field through the surface of the vessel is

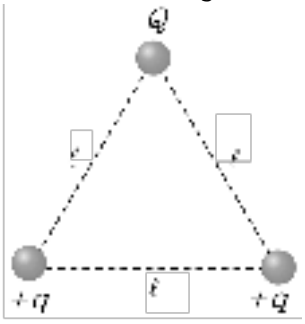
(1) Zero

(2) $\frac{q}{\varepsilon_0}$

(3) $\frac{q}{2\varepsilon_0}$

(4) $\frac{2q}{\varepsilon_0}$

23) Three charges Q,(+q) and (+q) are placed at the vertices of an equilateral triangle of side □ as shown in the figure. If the net electrostatic energy of the system is zero, then *Q* is equal to



(1) $\left(-\frac{q}{2}\right)$

(2) (–q)

(3) (+q)

(4) Zero

24) A transverse harmonic wave on a string is described by y = 3sin[36t + 0.018x + $\frac{\pi}{4}$] where x and y are in cm and t in s. The least distance between two sucessive crests in the wave is :-

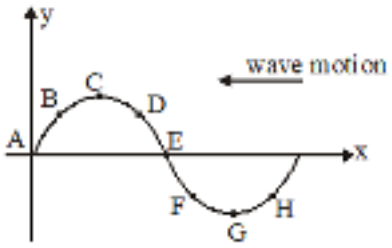
(1) 2.5 m

(2) 3.5 m

(3) 1.5 m

(4) 4.5 m

25) A transverse wave is travelling along a stretched string from right to left. The figure shown represents the shape of the string (snap short) at a given instant. At this instant



(1) the particles at A, B and H have downward velocity

(2) the particles at D, E and F have downward velocity

(3) the particles at C, E and G have zero velocity

(4) the particles at A and F have maximum velocity

26) A sound wave is passing through air column in the form of compression and rarefaction. In successive compression and rarefaction :-

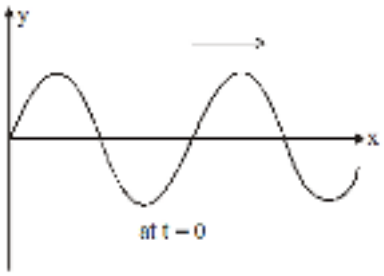
(1) pressure is constant

(2) density is constant

(3) position of the particles remains same

(4) No heat transfer

27) Figure shows a sinusoidal plot which describe the wave. Select the correct equation :-



(1) $y = A \sin(\omega t - kx)$

(2) $y = A \sin(kx - \omega t)$

(3) $y = A \sin(\omega t + kx)$

(4) $y = -A \sin(\omega t + kx)$

28) A heavy rope is suspended from a rigid support. A transverse wave pulse is set up at the lower end, then :

(1) the pulse will travel with uniform speed

(2) the pulse will travel with increasing speed

(3) the pulse will travel with decreasing speed

(4) the pulse cannot travel through the rope

29) The transverse displacement of a string (clamped at its both ends) is given by $y = 0.06\sin\left(\frac{2\pi}{3}x\right)\cos(120\pi t)$, where x and y are in m and t is in s. The length of the string is 1.5 m and its mass is 3.0×10^{-2} kg, then tension in string is :-

(1) 648 N

(2) 650 N

(3) 649 N

(4) 651 N

30) Two vibrating tuning forks produce progressive waves given by $y_1 = 4 \sin(500 \pi t)$ and $y_2 = 2 \sin(506\pi t)$. These tuning forks are held near the ear of a person. The person will hear

(1) 3 beats/s with intensity ratio between maxima and minima equal to 4

(2) 3 beats/s with intensity ratio between maxima and minima equal to 9

(3) 6 beats/s with intensity ratio between maxima and minima equal to 4

(4) 6 beats/s with intensity ratio between maxima and minima equal to 9

31) An organ pipe P_1 closed at one end vibrating in its first overtone. Another pipe P_2 open at both ends is vibrating in its third overtone. They are in a resonance with a given tuning fork. The ratio of the length of P_1 to that of P_2 is :

(1) $\frac{8}{3}$

(2) $\frac{3}{8}$

(3) $\frac{1}{2}$

(4) $\frac{3}{4}$

32) Two sitar strings, A and B, playing the note 'Dha' are slightly out of tune and produce beats of frequency 5 Hz. The tension of the string B is slightly increased and the beat frequency is found to decrease by 3 Hz. If the frequency of A is 425 Hz, the original frequency of B is :

(1) 430 Hz

(2) 420 Hz

(3) 428 Hz

(4) 422 Hz

33) A string of length 3 m and linear mass density 0.0025 kg/m is fixed at both ends. One of its resonance frequency is 252 Hz. The next higher resonance frequency is 336 Hz. Then the fundamental frequency will be

(1) 84 Hz

(2) 63 Hz

(3) 126 Hz

(4) 168 Hz

34) The ratio of the speed of sound in nitrogen gas to that in helium gas at 300 K is :

(1) $\sqrt{\frac{2}{7}}$

(2) $\sqrt{\frac{1}{7}}$

(3) $\frac{\sqrt{3}}{5}$

(4) $\frac{\sqrt{6}}{5}$

35) For a wave $y = y_0 \sin(\omega t - kx)$, for what value of λ is the maximum particle velocity equal to two times the wave velocity?

(1) πy_0

(2) $2\pi y_0$

(3) $\pi y_0/2$

(4) $4\pi y_0$

36) A cylindrical tube open at both ends has a fundamental frequency 400 Hz in air. The tube is dipped vertically in water so that half of it is in water. The fundamental frequency of air-column is

now :-

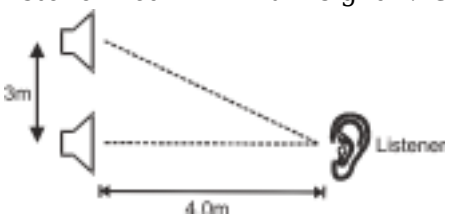
(1) 200 Hz

(2) 400 Hz

(3) 800 Hz

(4) 300 Hz

37) Two loudspeakers, as shown in fig. below separated by a distance 3 m, are in phase. Assume that the amplitudes of the sound from the speakers is approximately same at the position of a listener, who is at a distance 4.0 m in front of one of the speakers. For what frequency does the listener hear minimum signal ? Given that the speed of sound in air is 330 ms^{-1}.



(1) 165 Hz

(2) 300 Hz

(3) 400 Hz

(4) 600 Hz

38) **Assertion (A) :-** Sound travels faster on a rainy day than on a dry day.
Reason (R) :- With increase in humidity pressure increases.

(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false.

(4) **(A)** is false but **(R)** is true

39) **Assertion (A) :-** The change in air pressure has no effect on the speed of sound at constant temperature.
Reason (R) :- The speed of sound in a gas is directly proportional to pressure.

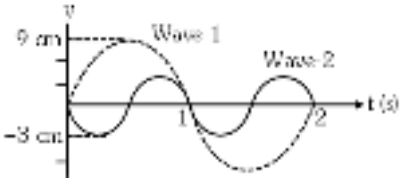
(1) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**.

(2) Both **(A)** and **(R)** are true and **(R)** is NOT the correct explanation of **(A)**.

(3) **(A)** is true but **(R)** is false

(4) **(A)** is false but **(R)** is true

40) Displacement- time graph for two waves, wave-1 and wave-2 are shown here. The ratio of the intensity of wave-1 to that of wave-2 is :



(1) 1 : 1

(2) 9 : 4

(3) 16 : 9

(4) 9 : 1

41) A wave is represented by $y = A\sin^2(kx - \omega t + \phi)$. The amplitude and wavelength of wave is given by

(1) $2A, \frac{2\pi}{k}$

(2) $A, \frac{2\pi}{k}$

(3) $\frac{A}{2}, \frac{2\pi}{k}$

(4) $\frac{A}{2}, \frac{\pi}{k}$

42) Two vibrating strings of the same material but length L and 2L have radii 2r and r respectively. They are stretched under the same tension. Both the strings vibrate in their fundamental modes, the one of length L with frequency ν_1 and other with frequency ν_2. Then ν_1/ν_2 is

(1) 1

(2) 1/2

(3) 3/2

(4) 2

43) In a resonance pipe the first and second resonance are obtained at depths 22.7 cm and 70.2 cm respectively. What will be the end correction?

(1) 1.05 cm

(2) 115.5 cm

(3) 92.5 cm

(4) 113.5 cm

44) A tunning fork produces 5 beats/sec when the length of a sonometer wire is either 1 m or 1.05 m. Calculate the frequency of tunning fork.

(1) 102.5 Hz

(2) 125 Hz

(3) 100 Hz

(4) 205 Hz

45) A string is clamped at both the ends and it is vibrating in its 4th harmonic. The equation of the stationary wave is
$y = 0.3 \sin(0.157x) \cos(200\pi t)$
The length of the string is (All quantities are in SI units.)

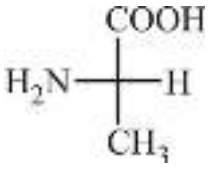
(1) 20 m

(2) 60 m

(3) 40 m

(4) 80 m

CHEMISTRY



1) What is relative configuration of following compound ?

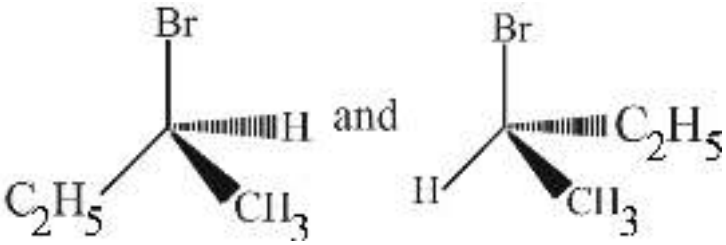
(1) D

(2) L

(3) cis

(4) trans

2) The given pair is :



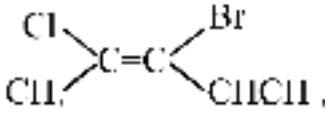
(1) Enantiomers

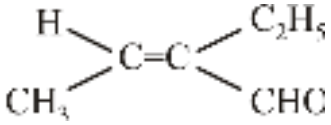
(2) Diastereomers

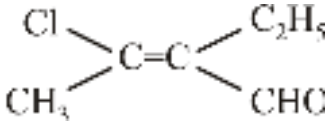
(3) Structural isomers

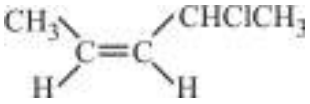
(4) Homomers

3) Which among the following has E configuration?

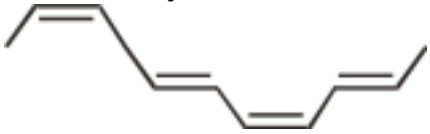
(1) 

(2) 

(3) 

(4) 

4) How many double bonds are having cis arrangement in following example :



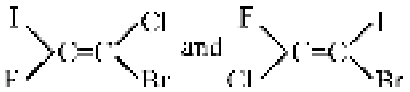
(1) One

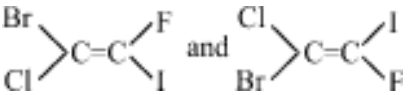
(2) Two

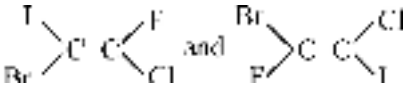
(3) Three

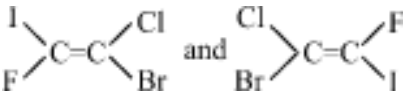
(4) Four

5) Which of the following pairs of compound are geometrical isomers :-

(1) 

(2) 

(3) 

(4) 

6) Find out the correct statement :-

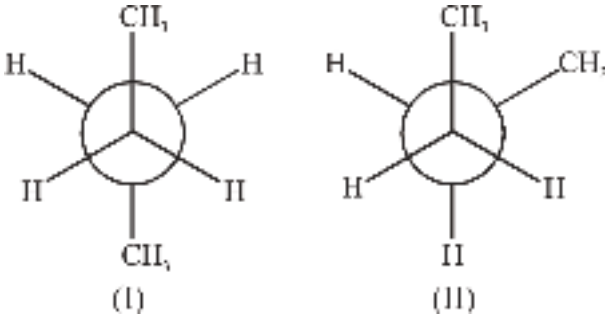
(1) In all the conformation bond angle and bond length remain same

(2) Melting point depends on packing efficiency

(3) If molecule rotate PPL in any direction, then it is considered as optically active

(4) All of these

7) Which of the following best explains the reason for the relative stabilities of the conformers shown?



(I) (II)

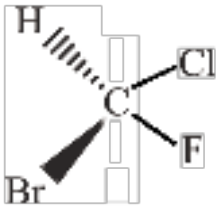
(1) I has more torsional strain

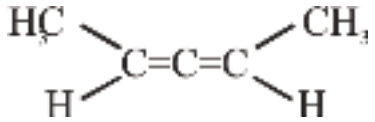
(2) I has less dihedral angle

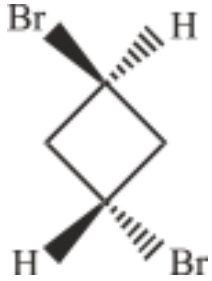
(3) II has more torsional and steric strain

(4) None of these

8) Which of the following has plane of symmetry :

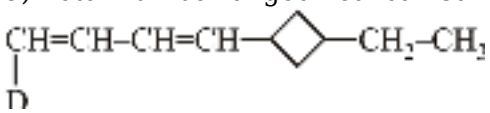
(1) 

(2) 

(3) 

(4) All

9) Total number of geometrical isomer in given structure will be ?



(1) 2

(2) 4

(3) 6

(4) 8

10) **Statement-I** :– Trihydroxyglutaric acid (COOH—CHOH—CHOH—CHOH—COOH) exists in four stereoisomeric forms, two of which are optically active while the other two are meso–forms.
Statement-II :– It contains two chiral and one pseudo–chiral carbon atom.

(1) Both Statement I and Statement II are correct.

(2) Statement I is correct but Statement II is incorrect.

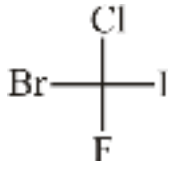
(3) Statement I is incorrect but Statement II is correct.

(4) Both Statement I and Statement II are incorrect.

11) Which of the following is enantiomer of given compound? 

(1) 

(2) 

(3) 

(4) None of these

12) How many structural isomers of alcohol with the formula C_4H_9OH are possible ?

(1) 4
(2) 3
(3) 5
(4) 6

13) Which of following is true statement about $\overset{\oplus}{C}H_3$.

(1) It has 8 electrons in valence shell
(2) Shape is tetrahedral
(3) Hybridisation of central atom is sp^2
(4) All are correct statements

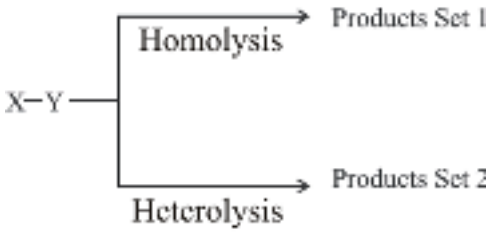
14) How many electrophiles are present in following group

BH_3, SO_3, NH_3, H_2O, NH_2–NH_2, $\overset{\ominus}{C}H_3$, CH_3–I

(1) 4
(2) 3
(3) 5
(4) 6

15) The correct decreasing order of –I effect for the following species :-
(a) $-NH_2$ (b) –Cl
(c) $-OCH_3$ (d) $-\overset{\oplus}{N}H_3$

(1) a > b > c > d
(2) d > c > b > a
(3) d > b > c > a
(4) d > a > b > c



16)
Products set(1) & products set (2) are :-

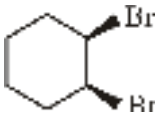
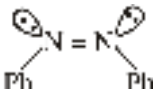
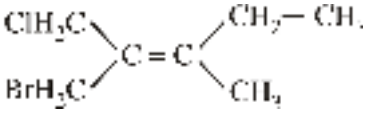
(1) Products set (1) - $\overset{\bullet}{X},\overset{\bullet}{Y}$ Products set (2) $\overset{\bullet}{X},\overset{\bullet}{Y}$

(2) Products set (1) $\overset{\oplus}{X},\overset{\ominus}{Y}$ - Product set (2) $\overset{\oplus}{X},\overset{\ominus}{Y}$

(3) Products set (1) $\overset{\bullet}{X},\overset{\oplus}{Y}$ - Product set (2) $\overset{\ominus}{X},\overset{\bullet}{Y}$

(4) Products set (1) $\overset{\bullet}{X},\overset{\bullet}{Y}$ - Products set (2) $\overset{\oplus}{X},\overset{\ominus}{Y}$

17) Match List-I with List-II and select the answer from the given codes.

List-I Structure of the compound		**List-II Name of the geometrical isomerism**	
(i)		(a)	Cis
(ii)		(b)	E-isomer
(iii)		(c)	Syn
(iv)		(d)	Trans

(1) i-a, ii-c, iii-b, iv-d

(2) i-a, ii-b, iii-c, iv-d

(3) i-b, ii-a, iii-c, iv-d

(4) i-c, ii-a, iii-d, iv-b

18) Chirality in molecule is refer for :-

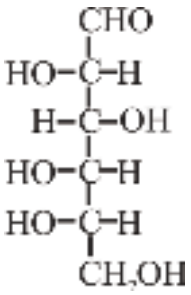
(1) Presence of chiral atom

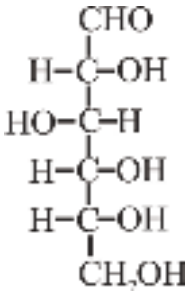
(2) Absence of chiral atom

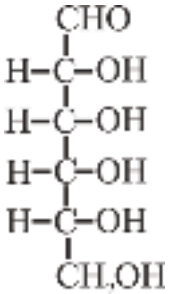
(3) Superimposability on mirror image

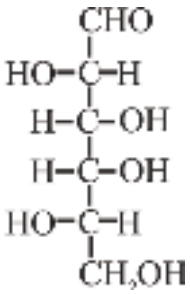
(4) Non-superimposability on mirror image

19) Which of the following is correct structure of L-glucose :-

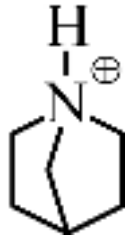
(1) 

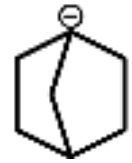
(2) 

(3) 

(4) 

20) Which of the following is stable

(1) 

(2) 

(3) 

(4) All of the above

21)

Which of the following stability order is not correct

(1) $\overset{\ominus}{C}H_3 < \overset{\ominus}{N}H_2$

(2) $CH_3O^{\ominus} < CH_3S^{\ominus}$

(3) $CH_3{-}\overset{\ominus}{C}H_2 < CH_2{=}CH^{\ominus}$

(4) $CH\equiv C^{\ominus} \leftrightarrow CH_2-CH^{\oplus}$

22) Total possible stereoisomer from given compound will be :- $CH_3-CH-CH(OH)-CH(OH)-CH_3$

(1) 6

(2) 8

(3) 4

(4) 10

23) Half life ($t_{1/2}$) and completion time (T) for a zero order reaction will be K = $\frac{mol}{L/sec}$ 0.001 and Initial concentration of reactant is 1 M

$A \rightarrow B$

K = Rate constant

(1) 500 min, 750 min

(2) 500 sec, 750 sec

(3) 500 sec, 1000 sec

(4) None of these

24) A hypothetical reaction $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given below :

$A_2 \rightleftharpoons 2A$ (fast)

$A + B_2 \rightarrow AB + B$ (Slow)

$A + B \rightarrow AB$ (fast)

The rate law expression should be :-

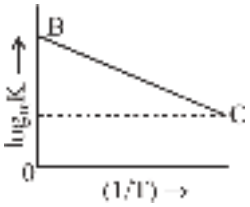
(1) $r = K'[A][B_2]$

(2) $r = K'[A_2]$

(3) $r = K'[A_2]^{1/2}[B_2]$

(4) None of these

25) Following figure shows a graph in $\log_{10}K$ vs $\frac{1}{T}$ where K is rate constant and T is temperature. The straight line BC has slope, $\tan\theta = -\frac{1}{2.303}$ and an intercept of 5 on Y–axis. Thus E_a, the energy of activation is :-



(1) 2.303 × 2 cal

(2) 2/2.303 cal

(3) 2 cal

(4) None of these

26) Two first order reactions have half-lives in the ratio 3 : 2. Calculate the ratio of time intervals t_1 : t_2 if t_1 is the time period for 25% completion of the first reaction and t_2 for 75% completion of the second reaction :-

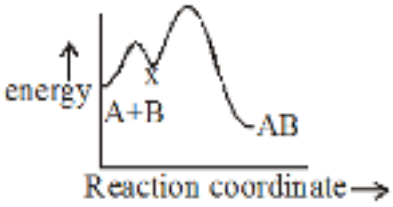
(1) 0.3 : 1

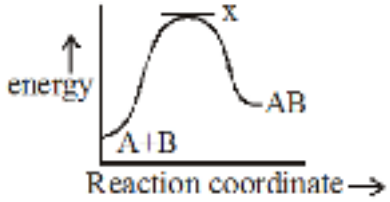
(2) 0.5 : 1

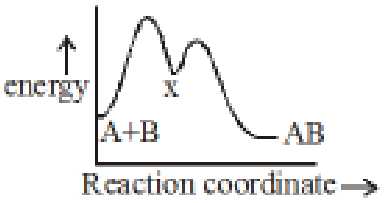
(3) 0.2 : 1

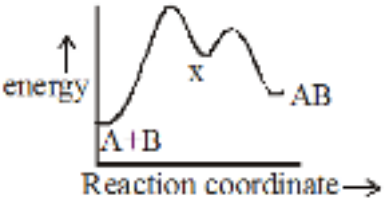
(4) 0.1 : 1

27) For an exothermic reaction; A+B → AB following two steps are involved :-
Step (i) A + B → X (slow)
Step (ii) X → AB (fast)
The progress of reaction can be best represented by :-

(1) 

(2) 

(3) 

(4) 

28)

For a first order reaction
$3A(g) \rightarrow 2B(g) + 3C(g) + D(g)$

if initial pressure of A is P_0 and after time 't' the pressure of gaseous mix is P_t the Rate Constant is :

(1) $K = \frac{1}{t}\ell n\frac{P_0}{2P_0 - P_t}$

(2) $K = \frac{1}{t}\ell n\frac{2P_0}{2P_0 - P_t}$

(3) $K = \frac{1}{t}\ell n\frac{P_0}{P_0 - P_t}$

(4) None of these

29) Given that
log2 = 0.3010, R = 2 × 10^{-3} Kcal mol^{-1} K^{-1}
A first order reaction is 50% completed in 20 minutes at 27ºC and in 10 minutes at 47ºC. The energy of activation of the reaction is :-

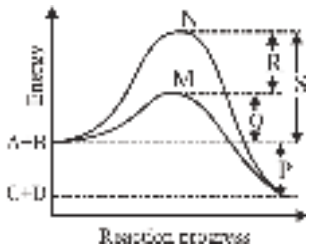
(1) 4.38 Kcal

(2) 3.43 Kcal

(3) 6.62 Kcal

(4) 1.32 Kcal

30) The curves M and N represent the variation of energy with reaction coordinate for the reaction



A(g) + B(g) → C(g) + D(g) in absence and presence of Inhibitors Of the curves M and N, which represents the E_a for the backward reaction in the presence of Inhibitors

(1) P

(2) Q

(3) S + P

(4) P + Q

31) For a zero order reaction, the plot of dissociated concentration (C_o – C_t) vs time is linear with :-
When
C_0 = initial concentration of reactant
C_t = concentration of reactant after time interval "t"

(1) +ve slope and zero intercept

(2) –ve slope and zero intercept

(3) +ve slope and non-zero intercept

(4) –ve slope and non-zero intercept

32) Rate of reaction increases 5.64 times with rise in temperature. If initial temperature is 25ºC and temperature coefficient of reaction = 2, find the final temperature. $(\sqrt{2} : 1.414)$

(1) 38°C

(2) 40°C

(3) 50°C

(4) 55°C

33) In the following first order reactions

(A) $\xrightarrow{K_1}$ Product, (B) $\xrightarrow{K_2}$ Product , find the ratio K_1/K_2 if 90% of (A) has been reacted in time 't' while 99% of (B) has been reacted in time 2t :-

(1) 1

(2) 2

(3) 1/2

(4) $2\left(\frac{1-2\log 3}{2-\log 11-2\log 3}\right)$

34) What will be the order of the reaction having following graph between $\log t_{1/2}$ and log a ?



(where a = initial concentration; $t_{1/2}$ = half life) :-

(1) Zero order

(2) First order

(3) Second order

(4) We can not predict the order by this graph

35) Consider the following statements :-
(i) Increase in concentration of reactant increases the rate of a zero order reaction
(ii) Rate constant k is equal to 'A' if $E_a = 0$
(iii) Rate constant k is equal to 'A' if $E_a = \infty$
(iv) $\log_e$ k vs T is a straight line
(v) $\log_e$k vs 1/T is straight line
where A = frequency factor
E_a = Activation energy
Correct statement are :-

(1) (i) and (iv)

(2) (ii) and (v)

(3) (iii) and (iv)

(4) (ii) and (iii)

36) If $K = Ae^{-\frac{E_a}{RT}}$ then
Statement-I : The pre-exponential factor A has the same units for all reactions.
Statement-II : $e^{-Ea/RT}$ has no unit.

(1) Statement-I is true, Statement-II is true ; Statement-II is correct explanation for Statement-I.

(2) Statement-I is true, Statement-II is true ; Statement-II is NOT a correct explanation for statement-I.

(3) Statement-I is true, Statement-II is false.

(4) Statement-I is false, Statement-II is true.

37) The activation energy of exothermic reaction
A → B is 80 KJ. The heat of reaction is 200 KJ. The activation energy for the reaction B → A will be-

(1) 80 KJ

(2) 120 KJ

(3) 280 KJ

(4) 200 KJ

38) Rate of a reaction can be expressed by Arrhenius equation as,
$K = Ae^{-E/RT}$
In this equation, E represents :

(1) the energy above which all the colliding molecules will react

(2) the energy below which colliding molecules will not react

(3) the total energy of the reacting molecules at a temperature, T

(4) the fraction of molecules with energy greater than the activation energy of the reaction

39) The rate of reaction at 273 K is R_0. The rate of reaction at 313 K will be
(Assuming temperature coefficient equal to 2) :-

(1) 16 R_0

(2) 64 R_0

(3) $R_0/32$

(4) $R_0/16$

40) The half-life period of a first order reaction is 15 minutes. The amount of substance left after one hour will be :

(1) $\frac{1}{4}$ of the original amount

(2) $\frac{1}{16}$ of the original amount

(3) $\frac{1}{32}$ of the original amount

(4) $\frac{1}{8}$ of the original amount

41)

Calculate mole fraction of solute of 1 molal aqueous solution of H_2SO_4?

(1) 0.0017

(2) 0.017

(3) 1.17

(4) 2.17

42) In a solution of 7.8 g benzene (C_6H_6) and 46.0 g toluene ($C_6H_5CH_3$), the mole fraction of benzene

is:-

(1) $\frac{1}{6}$

(2) $\frac{1}{5}$

(3) $\frac{1}{2}$

(4) $\frac{1}{3}$

43) What is the molarity of H_2SO_4 solution which is 98% by weight and the density of solution at 35°C is 1.84 gm/cm^3

(1) 4.18 M

(2) 8.14 M

(3) 18.4 M

(4) 18 M

44) The molarity of 20% (W/W) solution of sulphuric acid is 2.55 M. The density of the solution is :- [S = 32]

(1) 1.25 g cm^{-3}

(2) 0.125 g L^{-1}

(3) 2.55 g cm^{-3}

(4) Unpredictable

45) For a reaction $E_a = 0$ and $k = 3.2 \times 10^8 s^{-1}$ at 300K. The value of frequency factor at 400 K would be :-

(1) $6.4 \times 10^4 s^{-1}$

(2) $3.2 \times 10^4 s^{-1}$

(3) $3.2 \times 10^8 s^{-1}$

(4) $3.2 \times 10^5 s^{-1}$

BIOLOGY

1)

(a) Golgi complex is related to cell secretions.
(b) Lysosome is the part of endomembrane system.
(c) Mitochondria is associated with light reaction.
(d) Ribosome synthesize RNA.
In given statements which of the following are incorrect :-

(1) a and d

(2) b and d

(3) b and c

(4) c and d

2) **Assertion :** A few ions or molecules require ATP for their transport across the cell membrane.
Reason : The ions or molecules are transported across the membrane against their concentration gradient.

(1) Both Assertion & Reason are True & the Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not the correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

3) **Assertion :** The organisation of the prokaryotic cells are fundamentally similar even though prokaryotes exhibit a wide variety of shapes and functions.
Reason : All prokaryotes have a cell wall surrounding the cell membrane except in mycoplasma.

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true.

4) Which of the following statements are correct ?
(A) The number of chloroplast varies from 1 per cell of the Chlamydomonas, a green alga to 20-40 per cell in the mesophyll.
(B) Out of the two membranes of chloroplast and mitochondria, the inner membrane is relatively less permeable.
(C) Thylakoids are arranged in stacks like the piles of coins called grana (singular : granum).
(D) The stroma of the chloroplast contains enzymes required for the synthesis of carbohydrates and proteins.

(1) A, B & C only

(2) B, C & D only

(3) A, C & D only

(4) A, B, C & D all

5) If there are 24 chromosome in pollen mother cell of a flower then, how many chromosome would be present after Meiosis-I and Meiosis-II in the pollen respectively :

(1) 12, 24

(2) 12, 12

(3) 12, 6

(4) 24, 24

6)

Which of the following is/are incorrect with respect to reduction division?
(A) Diakinesis is marked by terminalization of chiasmata
(B) Splitting of centromere occurs at anaphase-I

(C) Pairing of homologous chromosomes occurs in prophase-I

(D) Interkinesis is short lived phase which is followed by prophase-II, a much simpler phase than prophase-I

(1) A, B

(2) B, D

(3) A, C

(4) Only B

7) Select the wrong pair :-

	Types of cells	Shape of cells
(1)	Red blood cells	Round and biconcave
(2)	Columnar epithelial cells	Long and narrow
(3)	Nerve cell	Unbranched and long
(4)	Mesophyll cells	Round and oval

(1) 1

(2) 2

(3) 3

(4) 4

8) Different cells have different sizes. Arrange the following in an ascending order of their size.
(i) Mycoplasma
(ii) Ostrich eggs
(iii) Human RBC
(iv) Bacteria

(1) i, ii, iii, iv

(2) i, iv, iii, ii

(3) ii, i, iii, iv

(4) iii, ii, i, iv

9) **Assertion :** Centrioles are essential for maintenance of continuity of cells in animals.
Reason : Centrioles give rise to spindle apparatus during cell division.

(1) Both Assertion & Reason are True & the Reason is the correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not the correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Assertion is False but Reason is True.

10) The .....A.....is the end of a chromosome that is elongated by an enzyme known as.....B.....
Sometimes a few chromosomes have non staining.....C.....at a constant location. This gives the appearance of a small fragment called the.....D.....
Fill in the blanks with most appropriate answer for A to D :-

(1) A-Kinetochore, B-Isomerase, C-Primary constriction, D-Telomeare

(2) A-Centromere, B-Telomerase, C-Secondary constrictions, D-kinetochore.

(3) A-Telomere, B-Hexokinase, C-Primary constrictions, D-Acrocentric

(4) A-Telomere, B-Telomerase, C-Secondary constrictions, D-Satellite

11) **Statement I :** The content of nucleolus remain continuous with rest of cytoplasm.
Statement II : Larger and more numerous nucleoli are present in cells actively carrying protein synthesis.

(1) Both statements I and II are correct

(2) Both statements I and II are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

12) In the following diagram, select the part correctly matched with its character or function.



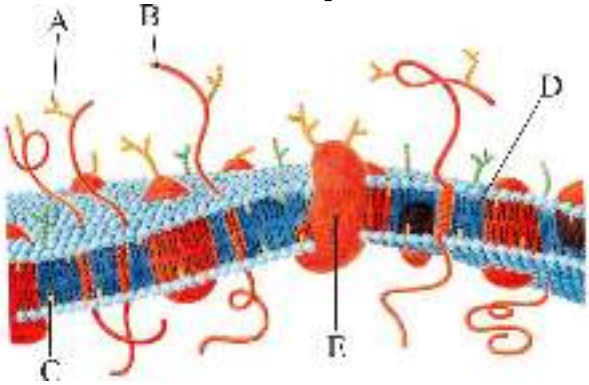
(1) A - The passage through which movement of RNA and protein molecules takes place in both direction.

(2) B - Frequently observed in cells actively involved in lipid synthesis and secretion

(3) C - Membrane bound structures composed of deoxyribonucleic acid and proteins.

(4) D - Provide surface for DNA and protein synthesis

13) The figure given below show the structure of plasma membrane, with its parts labelled from A to E, Identify the correct :-



(1) A → 52 percent of erythrocyte membrane

(2) B → Movement of this measured fluidity of membrane

(3) C → Help in faciliated transport of polar molecules

(4) D → Its polar tail contain saturated hydrocarbon

14) Read the statements carefully and select the correct option.
Statement-I : The size of typical bacterial cell is generally range from 1-3 μm while eukaryotic cells range from 10-200 μm.
Statement-II : In prokaryotes, ribosomes are about 15-20 nm in size and several ribosomes may attach to a single m-RNA and form a chain called polyribosomes.

(1) Both **Statement I** and **Statement II** are correct.

(2) Both **Statement I** and **Statement II** are incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) **Statement I** is correct but **Statement II** is incorrect.

15) In a container 2.5×10^{25} bacteria are reproducing by binary fission with rate of 25 minutes. How many bacteria will be produced after 1 hour 40 minutes.

(1) 40×10^{25}

(2) 24×10^{25}

(3) 16×10^{25}

(4) 4.4×10^{26}

16) In which one of the following options two events of cell cycle are not correctly matched with their particular phase ?

Events			Phase
(1)	(i)	Centriole duplication in the cytoplasm of animal cell	S
	(ii)	DNA replication in nucleus	
(2)	(i)	Splitting of centromere	Anaphase
	(ii)	Movement of centriole toward opposite poles of cell	
(3)	(i)	Activity of recombinase enzyme	Pachytene
	(ii)	Crossing over between non sister chromatid	
(4)	(i)	Terminalisation of chiasmata	Diakinesis
	(ii)	Nucleolus disappear	

(1) 1

(2) 2

(3) 3

(4) 4

17) Read the following four statements (A - D) :-
(A) The interphase lasts more than 95% of the duration of cell cycle.
(B) In animal cells during the s-phase, DNA replication begins in the nucleus and the centriole duplicates in the nucleolus
(C) Since the number of chromosomes in the parent and progeny cells is the same so mitosis is also

called as equational division.
(D) Mitosis usually results in the production of diploid daughter cells with identical genetic complement.
How many of the above statement/s is/are incorrect?

(1) One
(2) Two
(3) Four
(4) Three

18) Some events of mitotic cell division are given below in unordered manner -
(i) Condensation of chromatin is completed
(ii) Chromosome lose their identity as discrete elements
(iii) Sister chromatid separate
(iv) Initiation of the assembly of mitotic spindle between pair of centrioles
The correct order for steps of cell division is :-

(1) i → iv → iii → ii
(2) iv → i → iii → ii
(3) ii → i → iv → iii
(4) iv → iii → i → ii

19) A root tip cells has 14 chromosomes, how many chromosomes will the cell have, after splitting of centromere in mitotic anaphase ?

(1) 14
(2) 28
(3) 7
(4) 21

20) If mitotically dividing vegetative cell of flowering plant at G_1 stage has 20 chromosome with 20pg DNA then in anaphase cell will have

(1) 20 chromosomes/40pg DNA
(2) 40 chromosomes/40pg DNA
(3) 40 chromosomes/80pg DNA
(4) 10 chromosomes/40pg DNA

21) In which stage of mitosis, the chromosomes are composed of two chromatids ?

(1) Prophase & metaphase
(2) Anaphase and telophase
(3) Prophase and telophase
(4) Metaphase and anaphase

22) At anaphase - II of meiosis each chromosome contains:-

(1) 4 DNA

(2) 3 - DNA

(3) 2 - DNA

(4) 1 - DNA

23) Which of the following is/are correct with respect to mitosis ?
(A) Diakinesis is marked by appearance of chiasmata
(B) Duplication of centrioles occurs in Anaphase
(C) Sister chromatids of homologous chromosomes exhibit crossing over.
(D) In anaphase, centromeres splitand chromatids separate chromoatide move to opposite poles

(1) A & B

(2) B & D

(3) A & C

(4) Only D

24) Liquid endosperm in coconut results due to :-

(1) Karyokinesis followed by cytokinesis

(2) Failure of karyokinessis followed by cytokinesis

(3) Karyokinesis twice followed by single cytokinesis

(4) Karyokinesis is not followed by cytokinesis

25) **Statement I :** The plane of alignment of the chromosome at metaphase is referred to as the metaphase plate.
Statement II : In anaphase-I centromeres split and chromatids separate.

(1) Both statements I and II are correct

(2) Both statements I and II are incorrect

(3) Only statement I is correct

(4) Only statement II is correct

26) If 36 chromosome are present in a meiocyte then how many bivalent will arranged on metaphase-I?

(1) 36

(2) 72

(3) 18

(4) 9

27) Which event is **not** related with meiosis-I?

(1) Segregation of homologous chromosomes.

(2) Crossing over between non sister chromatids of homologous chromosomes.

(3) Separation of sister chromatids of a chromosome.

(4) Condensation of chromatin.

28) Events of mitosis is given in column-I and stages are given in column-II. Match the column-I and column-II and select the correct option given below :-

Column-I (Events)		**Column-II (Stages of mitosis)**	
A	Centriole duplication	(i)	G_0
B	Most of cell organelles duplicate	(ii)	G_2
C	Tubulin protein synthesis	(iii)	S
D	Cells do not appear to exhibit division	(iv)	G_1

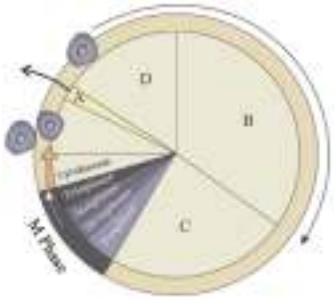
(1) A-ii, B-iv, C-iii, D-i

(2) A-iii, B-iv, C-ii, D-i

(3) A-iv, B-ii, C-iii, D-i

(4) A-iii, B-iv, C-i, D-ii

29) Given below is diagrammatic view of cell cycle indicating formation of two cell from one cell.



Mark the correct one for the events of this cell cycle :-

(1)	A	G_0	Cells is metabolically inactive
(2)	B	S	Amount of DNA per cell doubles but chromosome number remain same
(3)	C	G_2	Interval between mitosis and initiation of DNA replication where cell continuously grows
(4)	D	G_1	Shortest phase of cell cycle in which most of the organelles duplicate

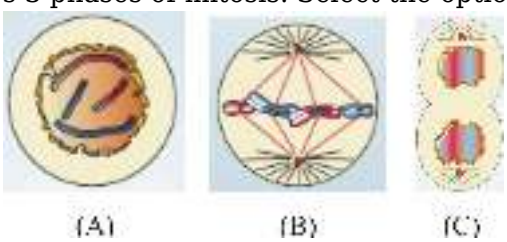
(1) 1

(2) 2

(3) 3

(4) 4

30) The figures given below shows 3 phases of mitosis. Select the option giving correct identification together with the correct event ?



(1) C-Telophase-Nuclear envelope assembles around the chromosome clusters.

(2) B-Anaphase-Segregation of homologous chromosomes.

(3) A-Prophase- synapsis is completed.

(4) C-Metaphase-Condensation of chromatin to form chromosome.

31) Identify the correct match from the Column-I and Column-II and Column-III.

	Column-I		**Column-II**		**Column-III**
1	Biodiversity	a	used in natural classification	i	Study of both external and internal type of characters
2	Gross morphology	b	identification, Nomenclature and classification	ii	Different type of plants and animals
3	Systematics	c	includes cytological informations	iii	Evolutionary relationships
4	Cytotaxonomy	d	different types and number of species	iv	Use in solving taxonomic problem

(1) 1-d,ii 2-a,i 3-b,iii 4-c,iv

(2) 1-a,i 2-b,ii 3-c,iii 4-d,iv

(3) 1-d,iv 2-c,iii 3-b,ii 4-c,i

(4) 1-d,iii 2-c,ii 3-a, iv 4-b, i

32) List some of the organisms is given below :-
Chlorella, Riccia, Pisum, Chlamydomonas, Cycas, Pteridium, Capsella, Gonyaulax, Euglena, Amoeba, Physarum, Fucus, Chara.
How many are considered in kingdom protista according to R.H. Whittaker?

(1) Seven

(2) Eight

(3) Nine

(4) Six

33) How many of the following statements are correct:-
(A) Protists reproduce asexually and sexually and have flagella or cilia.
(B) Slime moulds are saprophytic protists.
(C) The cell wall of dinoflagellates has stiff silicated plates on outer surface.
(D) Diatoms float passively with water currents
(E) The pigments of euglenoids are identical to those present in higher plants.

(1) Five

(2) Four

(3) Three

(4) Two

34) A collection of species, which bears a close resemblance to one another in the morphological characters of the floral parts is known as :-

(1) Family

(2) Genus

(3) Division

(4) Variety

35) Choose the correct from followings :-

(1) *Mangifera*, is a name of species

(2) Polymoniales, is a name of family

(3) Dicotyledonae, is a name of division

(4) *Triticum*, is a name of genus

36) How many are the correct statements from the following?
(A) Each taxonomic category is a unit of classification
(B) Taxonomic groups are distinct biological entities
(C) Genera are aggregates of closely related families
(D) Both vegetative and reproductive characters are used to characterise the families.

(1) Two

(2) Three

(3) One

(4) Four

37)

Character	**Monera**	**Protista**	**Fungi**
Body organisation	Cellular	Cellular	(A)
Cell wall	Present	(B)	Present
Sexual reproduction	(C)	Present	Present

What are (A) (B) & (C) in the above given table ?

	(A)	**(B)**	**(C)**
(1)	Unicellular or multicellular	Absent	Absent
(2)	Multicellular or loose tissue	Present in some members	Absent

(3)	Multicellular or loose tissue	Present in all members	Absent
(4)	only unicelluar	Absent in all members	Present in some members

(1) 1
(2) 2
(3) 3
(4) 4

38) Choose the correct match :-

(1) Mesosomes - Cell respiration
(2) Plasmid - bear photosynthetic pigments
(3) Flagella - help in formation of conjugation tube
(4) Capsule - help in locomotion

39) Choose the incorrect statement from following :-

(1) Ribosomes are the site of protein synthesis
(2) Pili do not play a role in motility
(3) Inclusion bodies are bounded by membrane
(4) Cell wall prevent the bacterium from bursting and collapsing

40) The speed of binary fission in *Escherichia coli* is :-

(1) Considerably slower than mitosis of advanced organism since bacteria are primitive
(2) Considerable faster that mitosis of advanced organism
(3) Essentially the same speed as mitosis in advanced organism
(4) Considerably slower than meiosis of advanced organism

41) ***Assertion (A)*** :- Bacteria show the most extensive metabolic diversity.
Reason (R) :- The vast majority of bacteria are chemo-autotrophs/chemosynthetic autotrophs.

(1) Both A and R are correct and R is the correct explanation of A.
(2) Both A and R are correct but R is not the correct explanation of A.
(3) A is True but R is False.
(4) R is True but A is False.

42) Read the following terms carefully :-
Cilia, Flagella, Heterocyst, Unicellular, Eukaryotes, Photosynthesis, Embryo, Zygote, Tissue, Seed, Murein, Pili, Pellicle, Chemosynthesis, Nitrogen fixation.
How many terms are only and only related with Monera and not found in any other kingdom.

(1) 8
(2) 7

(3) 5

(4) 10

43) Consider the following four statements (A-D) and select the option which includes all the correct ones only.
(A) Archaebacteria differ from other bacteria in having a different cell wall structure
(B) Some BGA can fix atmospheric nitrogen in specialised cells called heterocysts
(C) All Eubacteria have chlorophyll 'a' pigment similar to green plants.
(D) Mycoplasma can survive without oxygen.

(1) Statements B, C & D

(2) Statements A, C only

(3) Statements B, D only

(4) Statements A, B & D

44) In ascomycetes and basidiomycetes, the ascospores and basidiospore are produced respectively :-

(1) Exogenously and endogenously

(2) Endogenously and endogenously

(3) Exogenously and exogenously

(4) Endogenously and exogenously

45) Basidiocarp and ascocarp are the fruiting bodies of fungi. They can be seen in :-

(1) *Alternaria & Agaricus* respectively

(2) *Puccinia & Ustilago* respectively

(3) *Claviceps & Agaricus* respectively

(4) *Agaricus & Claviceps* respectively

46) Consider the genera given below and answer the question asked next to them :-
Chlorella, Sargassam, Ulothrix, Laminaria, Ectocarpus, Volvox, Polysiphonia, Porphyra, Fucus, Gelidium, Dictyota, Acetabularia
How many above genera belong to red algae and brown algae respectively ?

(1) Six and Five

(2) Three and Five

(3) Three and Four

(4) Five and Six

47) Red tide is caused by :-

(1) Rapid multiplication of red algae

(2) Rapid multiplization of euglenoids

(3) Rapid multiplication of slime molds

(4) Rapid multiplication of dinoflagellates

48) Which of the following is not a correct statement?

(1) Euglenoids have a protein rich layer instead of cell wall
(2) Slime moulds are decomposer protists
(3) Cell wall of dinoflagellates is indestructible
(4) Citrus canker diseases caused by bacteria

49) A group of college student orderd a pizza with pineapple, onion and mushroom. Mushroom belong to which kingdom ?

(1) *Agaricus*
(2) Ascomycetes
(3) Fungi
(4) Basidiomycetes

50) Match the column-I with column-II and choose the correct option :-

	Column-I		**Column-II**
A	Deuteromycetes	(i)	Gametes are similar in morphology
B	Ascomycetes	(ii)	Sexual spores are produced endogenously
C	Basidiomycetes	(iii)	Imperfect fungi
D	Phycomycetes	(iv)	Sex organ are absent

(1) A-iii, B-iv, C-i, D-ii
(2) A-iv, B-i, C-iii, D-ii
(3) A-iv, B-iii, C-i, D-ii
(4) A-iii, B-ii, C-iv, D-i



51) How many statement is/are **correct** regarding given diagram ? (a) This virus have helical symmetry
(b) This virus have genetic material RNA
(c) This virus is smallar than prions
(d) This virus discover by pasteur (e) This virus have spherical shape

(1) 2
(2) 3
(3) 4
(4) All

52) Read the following statements :-
(A) These algae are marine with greater concentration found in warmer areas in both well lighted region close to surface of water and at greater depth in oceans.

(B) Thallus is mostly multicellular having complex body organisation
(C) They reproduce asexually by non motile spore and sexually by non motile gametes
(D) Sexual reproduction is oogamous and accomplaned by complex post fertilisation development.
The above statements reflects that the plant growp would be :-

(1) Brown algae

(2) Red algae

(3) Green algae

(4) Golden algae

53) The given table shows classes of algae and their main characteristics. Some characteristics in this table mentioned as A, B, C and D, Observe the table and choose the option in which A, B, C and D are correctly identified :-

Classes	**Chlorophyceae**	**Phaeophyceae**	**Rhodophyceae**
Major Pigments	Chl a, b	Chl a, c	**A**
Stored Food	Starch	**B**	Floridean starch
Cell Wall	**C**	Cellulose and algin	Cellulose, pectin, polysulphate ester
Flagellar Number	2-8	2	**D**

	A	**B**	**C**	**D**
(1)	Chl a, e	Fat & oil	Peptidoglycan	Many
(2)	Chl a, d	Mannitol, Laminarin	Cellulose	Absent
(3)	Chl a, d	Paramylum	Lipoprotein	One
(4)	Chl a, c	Glycogen	Cellulose	Absent

(1) 1

(2) 2

(3) 3

(4) 4

54) Read the statement about bryophytes and arrange them in sequential order
(A) Zygote produce multicellular body called sporophyte
(B) Antherozoid released into water
(C) Antherozoid fuse with egg to produce zygote
(D) Antherozoid come in contact with archaegonia

(1) C→D→A→B

(2) B→D→A→C

(3) C→D→B→A

(4) B→D→C→A

55) Which of the following statement is correct ?

(1) In *Sphagnum*, sexual reproduction takes place by green, multicellular buds called gemmae

(2) The sporophyte of bryophytes is differentiated into foot, seta and capsule and it is a free-living generation

(3) Protonema is a green, creeping, branched and filamentous stage in mosses, which develops directly from a spore.

(4) Liverworts have an more elaborate mechanism of seed dispersal, as compare to mosses

56) Which option is not related with Pteridophytes ?

(1) Free living gametophyte

(2) Mainly Monoecious sporophytes.

(3) Precursor of seed habit

(4) Zygote produces a multicellular well-differentiated sporophyte

57) Which statement is correct for pteridophyta ?

(1) Gametophyte is autotrophic, while sporophyte is parasitic.

(2) Sporophyte is heterotrophic and gametophyte is autotrophic.

(3) Both gametophyte and sporophyte are usually autotrophic.

(4) Only sporophyte is autotrophic, while gametophyte is parasite.

58) In a plant the body is differentiated into root, stem and leaf, embryo is formed, vascular tissues are present and haploid endosperm is formed. What is a most important character that includes this plant as a member of gymnosperm?

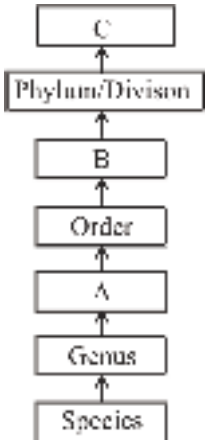
(1) Endosporic germination of spore.

(2) Formation of pollen tube for fertilization

(3) Ovules are naked

(4) Pollination by wind

59) The given flowchart represents the hierarchy of various taxonomic categories.
Identify the missing categories (A,B and C) and select the correct statements regarding these.



(i) A is the taxonomic category which contains a number of related genera.
(ii) Examples of category B are Monocotyledoneae, Dicotyledoneae, Mammalia, etc.
(iii) C represents the basic unit of taxonomic hierarchy.
(iv) Examples of category C are Fungi, Monera, Protista etc.

(1) (i) and (ii) only

(2) (iii) and (iv) only

(3) (i), (ii) and (iv) only

(4) (i), (ii), (iii) and (iv)

60) Arrange the following different taxa in ascending sequence according to their rank.
A. Poaceae B. Polymoniales
C. Panthera D. Monocotyledonae

(1) C – A – B – D

(2) D – C – B – A

(3) C – B – A – D

(4) D – B – A – C

61) A student observed a plant of chinarose and observe following characters in this plant. Alternate phyllotaxy, parallel venation, twisted aestivation of petals, diadelphous stamens, epiphyllous stamens, axile placentation, epigynous flower and superior ovary. Which of the above-mentioned characters are wrongly identified by the student ?

(1) Parallel venation, diadelphous stamens, axile placentation and epigynous flower

(2) Parallel venation, diadelphous stamens, axile placentation and superior ovary

(3) Parallel venation, diadelphous stamens, epiphyllous stamens and epigynous flower

(4) Parallel venation, diadelphous stamens, epiphyllous stamens and superior ovary

62) Floral formula a flower is given below. Identify the statements which are true for the flower, summarised in floral formula.: $\oplus \male\female K_{(5)} \overbrace{C_{(5)}\ A_5}\ \underline{G_{(2)}}$ A. Cohesion and adhesion of stamens both are present.
B. Bicarpellary syncarpous gynoecium with superior ovary is present
C. Placentation and aestivation both can be identified by this floral formula
D. This formula shows that flower is actinomorphic and complete

(1) A and C

(2) B and D

(3) A and D

(4) A and B

63) Match the column I with column II and select the correct option rom options given below:

	Column-I		**Column-II**
A	Parietal	G	Mustard
B	Free central	H	*Dianthus*
C	Axile	I	Chinarose
D	Basal	J	Marigold

(1) A:G; B:I; C:H; D:J

(2) A:G; B:H; C:I; D:J

(3) A:G; B:I; C:J; D:H

(4) A:G; B:H; C:J; D: I

64) **Statement A :** All unisexual flowers are incomplete flower
Statement B : All bisexual flowers are complete

(1) Statement A and statement B both are true

(2) Statement A and B both are false

(3) Only statement B is true

(4) Only statement A is true

65) **Assertion(A):** The angiosperms show large diversity in morphology and are all characterized by presence of roots, stems, leaves, flowers and fruits.
Reason (R): Various morphological features are used to describe a flowering plant beginning with its habit and then floral characters.

(1) A and R both are true and R explain A

(2) A is true but R is false

(3) A and R both are false

(4) A and R both are true and R does not explain A

66) A monocot plant can be differentiated from dicot plant in having ?

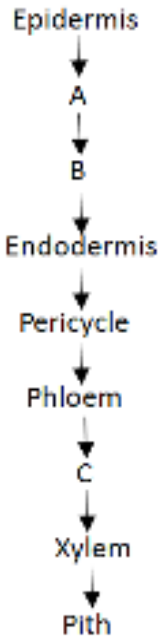
(1) Parallel venation

(2) Tetramerous flowers

(3) Reticulate venation

(4) Embryonal axis

67) Different structures present in internal structure of dicot stem from outer to inner side is given in flow chart. In the given flow chart identify the A, B and C and find out the correct statements



A. A and B are parenchymatous tissue
B. A and B are the part of cortex and B has conspicuous intercellular spaces
C. C is intra fascicular cambium and it is a primary meristem
D. A, B and C are living tissues

(1) B and C only
(2) B, C and D
(3) A, B and D are correct
(4) B and D only

68) Which of the following is not associated with mustard?

(1) Alternate phyllotaxy of leaf
(2) Tap root system
(3) Valvate aestivation of corolla
(4) Staminode

69) Ephiphyllous stamens are present in :

(1) Pea
(2) China rose
(3) Mustard
(4) Lily

70) Which of the following has adventitious root meant for food storage?

(1) Sweet potato
(2) Turnip
(3) Carrot
(4) Banyan Tree

71) The region of root tip where root hairs are present is known as :

(1) Region of Meristematic activities
(2) Region of Elongation
(3) Region of differentiation
(4) Root cap region

72) Many simple leaves are developed on one node in :

(1) Guava
(2) *Alstonia*
(3) Mustard
(4) *Calotropis*

73) Axillary bud is present in axil of:

(1) Simple leaf only
(2) Pinnately compound leaf only
(3) Palmately compound leaf only
(4) Simple and compound leaf both

74) Which of the following is not a function of stem?

(1) Absorption of water and minerals from soil
(2) Storage of food
(3) Conduct water and minerals from root to leaf
(4) Mode of vegetative propagation

75) Edible part of mango and coir of coconut is respectively :

(1) Mesocarp and Mesocarp
(2) Mesocarp and endosperm
(3) Epicarp and endosperm
(4) Endocarp and endosperm

76) Statement A: Polyadelphous condition of stamens is present in lemon
Statement B: Variation in length of stamen filaments are present in mustard and *Salvinia*

(1) Statement A and B both are true
(2) Only statement B is true
(3) Only Statement A is true
(4) Statement A and B both are false

77) Match the column-I with Column-II and select the correct option from the options given below

	Column-I		**Column-II**
A	Parenchyma	P	Hypodermis of monocot stem
B	Collenchyma	Q	Endodermis of root

C	Sclerenchyma	R Hypodermis of dicot stem
		S Bundle sheath of monocot leaf
		T Bundle sheath of monocot stem
		U Bundle cap
		V Pith of root

(1) A: Q, V; B: R; C: P, S, T, U

(2) A: Q, V; B: R, S; C: P, T, U

(3) A: Q, S, V; B: R; C: P, T, U

(4) A: Q, V; B: R, U; C: P, S, T

78) Which of the following is not related to vascular bundle in monocot root?

(1) Radial vascular bundle

(2) Xylem and phloem are present in different radii

(3) Xylem is exarch

(4) Vascular bundles are scattered

79) Assertion (A): Size of vascular bundles are variable in leaves of dicot plants
Reason (R): Reticulate venation is present in dorsiventral leaves of dicot plants

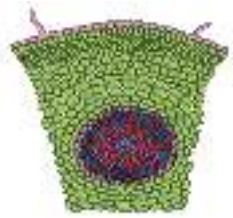
(1) A and R both are true and R explain A

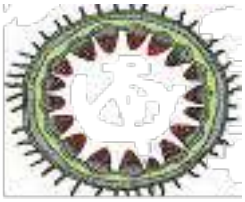
(2) A is true but R is false

(3) A and R both are false

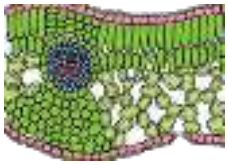
(4) A and R both are true and R does not explain A

80) Which of the following is not related to sunflower?

(1) 

(2) 

(3) 

(4) 

81) **Assertion (A):** The internal structure of plants can be used to differentiate monocot stem from dicot stem
Reason (R): Monocotyledonous stem and dicotyledonous stem show marked variation in their internal structures

(1) A and R both are true and R explain A

(2) A is true but R is false

(3) A and R both are false

(4) A and R both are true and R does not explain A

82) The ground tissue system in dicot stem is differentiated in :

(1) Cortex, pericycle and pith

(2) Cortex, pericycle and vascular bundles

(3) Cortex and pith only

(4) Cortex and stele

83) Match the column I with column II and select the correct option from options given below

	Column-I		**Column-II**
A	More than six radial vascular bundle	P	Monocot root
B	Bulliform cells	Q	Monocot leaf
C	Reticulate venation	R	Dicot leaf
D	Inconspicuous pith	S	Dicot root

(1) A: S; B: Q; C: R; D: P

(2) A: P; B: Q; C: R; D: S

(3) A: S; B: R; C: Q; D: P

(4) A: P; B: R; C: Q; D: S

84) Monocot root can be differentiated from monocot stem in having:

(1) Collenchymatous hypodermis

(2) Casparian strips in endodermal cells

(3) Large number of radial vascular bundles

(4) 2 and 3 both

85) Read the following statements regarding internal structure of flowering plants and select the correct statements and select the correct option from options given below:
a. The guard cells regulate the opening and closing of stomata
b. In the monocotyledons plants the vascular bundles have no cambium present in them.
c. The cells of root epidermis protrude in the form of multicellular root hairs
d. The innermost layer of the cortex in stem is called the endodermis and have suberized Casparian strips**.**

(1) a and b only

(2) a and c only

(3) a, b and c
(4) a, b and d

86) Which of the following is true regarding trichome?

(1) They are multicellular usually
(2) They are usually present in monocot stem
(3) They are present in dicot root
(4) They are always soft and hair like

87) All the layers present in between epidermis to pericycle in dicot stem is known as:

(1) Endodermis
(2) Stele
(3) Cortex
(4) Pith

88) Cohesion and adhesion of stamens both are observed in :

(1) China rose
(2) Pea
(3) Lily
(4) Mustard

89) Which of the following is correct floral diagram of mustard flowers?

(1) 

(2) 

(3) 

(4) 

90) Statement A: Stomata are more in number on adaxial epidermis in dorsiventral leaves of dicot plants
Statement B: Leaf mesophyll in dorsiventral leaves is differentiated in palisade and spongy parenchyma

(1) Statement A and B both are correct

(2) Only statement A is correct

(3) Only statement B is correct

(4) Statement A and B both are incorrect

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	1	1	1	3	1	3	2	2	4	2	2	3	3	4	1	4	3	3	2	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	1	2	2	4	2	2	1	2	2	2	1	3	1	2	1	3	3	2
Q.	41	42	43	44	45															
A.	4	1	1	4	4															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	1	3	2	4	4	3	3	4	1	2	1	3	2	3	4	1	4	1	4
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	2	3	3	3	1	3	1	3	3	1	3	1	1	2	4	3	2	1	2
Q.	86	87	88	89	90															
A.	2	1	3	1	3															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	1	2	4	2	4	3	2	1	4	4	1	2	3	1	2	1	2	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	1	4	4	4	3	3	3	2	2	1	1	4	2	2	4	2	2	1	3	2
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	3	3	4	4	4	2	4	3	3	4	1	2	2	4	3	2	3	3	3	1
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	2	2	4	4	1	2	4	4	1	3	2	4	1	1	3	3	4	1	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	1	2	4	1	1	3	1	2	3										

SOLUTIONS

PHYSICS

1) **Explain the Question :** Find which surface has maximum electric flux by comparing enclosed charge and writing flux for each.
Concept : This question is based on Gauss's law.
Solution :
$S_1 : q_{S_1} = q_1 + q_2 = 1 + 2 = +3\ \mu C$
$S_2 : q_{S_2} = q_2 + q_3 = 2 - 3 = -1\ \mu C$
$S_3 : q_{S_3} = q_1 + q_2 + q_3 = 0$
Flux through each

(Gauss's law $\phi = \frac{q_{enclosed}}{\epsilon_0}$) :

Flux through S_1: $\phi_{S_1} = \frac{3\mu C}{\epsilon_0}$

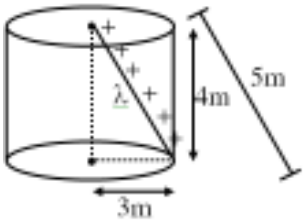
Flux through S_2: $\phi_{S_2} = \frac{-1\mu C}{\epsilon_0}$

Flux through S_3: $\phi_{S_3} = 0$

Maximum flux is through S_1
Final Answer: (1)

2)

Explain the Question : We have to calculate flux



passing through the cylinder
Concept : This question is based on flux

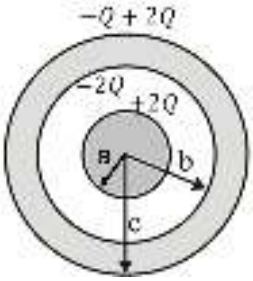
$(\phi) = \frac{q_{inside}}{\epsilon_0}$

Solution : total charge enclosed inside the cylinder
$q = \lambda \times 5$
[$\lambda = 8.85\ \mu C/m$]

$\phi = \frac{q}{\varepsilon_0} = \frac{\lambda \times 5}{\varepsilon_0} = \frac{8.85 \times 10^{-6} \times 5}{8.85 \times 10^{-12}}$

$\phi = 5 \times 10^6$ V-m
Final answer : option (1)

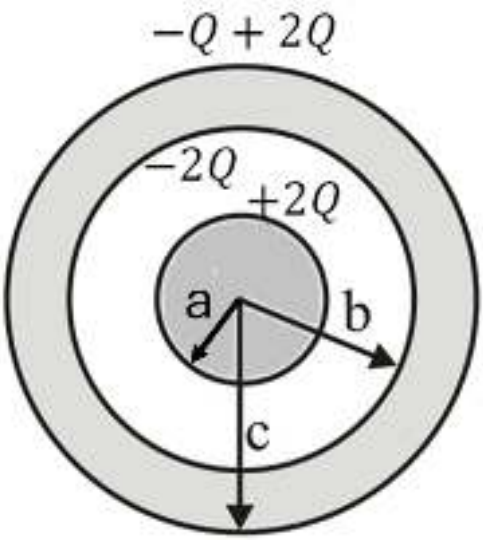
3) **Question Explanation:**



we have to calculate surface charge density on inner and outer surface of spherical shell , according to the charge distribution shown.

Concept:
Redistribution of charges for conductors.

Solution:



∵ surface charge density is given by

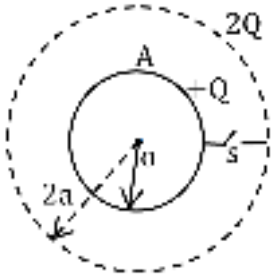
$$\sigma = \frac{Q}{4\pi R^2}$$

then,

$$\sigma_{inner} = \frac{-2Q}{4\pi b^2} \; ; \; \sigma_{outer} = \frac{Q}{4\pi c^2}$$

Final Answer: Option(1).

4) **Explain Question:**



If initial potential of point 'A' is 'V' we have calculate new potential at 'A' when switch is closed.

Concept: Electric potential in case of concentric conductors.

Solution: be fore the losing of switch

$$V_A = \frac{KQ}{a} + \frac{K(2Q)}{2a} = \frac{2KQ}{a} = V ----(1)$$

After closing switch the charge 'q' will flow to other surface.

So, $V_A' = \frac{K(3Q)}{2a} \quad ---(2)$

$$\frac{V_A'}{V} = \frac{\frac{K(3Q)}{2a}}{\frac{2KQ}{a}} \Rightarrow \frac{V'}{V} = \frac{3}{4}$$

$$\frac{V_A'}{V} = \frac{3}{4}V$$

Final Answer: Option (3)

5) $V^0 = \frac{kQ_{net}}{R}$

$V^0 = \frac{k(2Q)}{R}$

6) $E = -\frac{dV}{dx}$

E ∝ slope of V – x graph

⇒ $E_4 > E_2 > E_1 = E_3$

7) $v_1 = \frac{\lambda}{4\pi\in_0}(\pi) = \frac{\lambda}{4\in_0}$

$v_2 = \frac{\lambda}{4\pi\in_0}\left(\frac{\pi}{2}\right) = \frac{\lambda}{8\in_0}$

$v_3 = \frac{\lambda}{4\pi\in_0}\left(\frac{\pi}{3}\right) = \frac{\lambda}{12\in_0}$

$v_1 > v_2 > v_3$

8)

Question Explanation : We need to find the change on the surface x , for the system of three conducting plates shown.

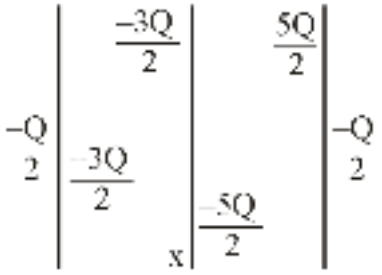
Concept : Redistribution of change in case of conductors (and properties of conductors)

Solution : Charge on leftmost and rightmost surface after redistribution

$$q = \frac{Q + 2Q + (-4Q)}{2} = \frac{-Q}{2}$$

Final charges after redistribution will be

$\Rightarrow x = \frac{-3Q}{2}$

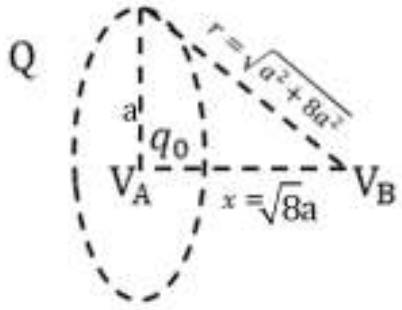


Final answer : Option (2)

9) $W_1 = -e[70\ V - 100\ V] = +30\ eV$
$W_2 = -e[70\ V - 90\ V] = +20\ eV$
$W_3 = -e[70\ V - 80\ V] = +10\ eV$
$W_4 = -e[80\ V - 100\ V] = +20\ eV$
$\Rightarrow\ W_1 > W_2 = W_4 > W_3$

10) $V_A = V_C$
$c = a + b$

11) **Explain Question:**
We have to calculate W.D in moving a charge q_0 from centre of ring to an arial point $x = \sqrt{8a}$:



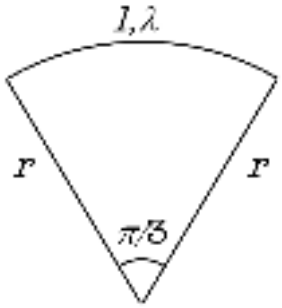
Concept:
work done $= q[V_B - V_A]$

Solution:
$$V_B = \frac{KQ}{r} = \frac{KQ}{\sqrt{a^2 + 8a^2}} = \frac{KQ}{3a}$$
$$V_A = \frac{KQ}{a}$$
$$W = q_0[V_B - V_A]$$
$$W = q_0\left[\frac{KQ}{3a} - \frac{KQ}{a}\right]$$
$$W = q_0\left[-\frac{2}{3} \times \frac{1}{4\pi\varepsilon_0} \times \frac{Q}{a}\right]$$
$$W = \frac{-Qq_0}{6\pi\varepsilon_0 a}$$

Final answer: option (2)

12) $\Delta V = E r \cos\theta$
$\Delta V_{max} = Er = 10 \times 2 = 20V$
So $\Delta V < 20$ Volt
Hence, option (3)

13) Length of the arc $= r\theta = \frac{r\pi}{3}$



Charge on the arc $= \frac{r\pi}{3} \times \lambda$

 Potential at center $= \frac{kq}{r}$

$= \frac{1}{4\pi\varepsilon_0} \times \frac{r\pi}{3}\frac{\lambda}{r} = \frac{\lambda}{12\varepsilon_0}$

14) Charge q will momentarily come to rest at a distance r from charge Q when all it's kinetic energy converted to potential energy i.e. $\frac{1}{2}mv^2 = \frac{1}{4\pi\varepsilon_0}.\frac{qQ}{r}$
Therefore the distance of closest approach is given by
$r = \frac{qQ}{4\pi\varepsilon_0}.\frac{2}{mv^2} \Rightarrow r \propto \frac{1}{v^2}$
Hence if v is doubled, r becomes one fourth.

15) P.D between two shells is independent of charge on outer shell.

16) $E_x = -\frac{dV}{dx} = -(6 - 8y^2), E_y = -\frac{dV}{dy} = -(-16xy - 8 + 6z)$
$E_z = -\frac{dV}{dz} = -(6y - 8z)$
At origin x = y = z = 0 so, $E_x = -6, E_y = 8$ and $E_z = 0$
$\Rightarrow E = \sqrt{E_x^2 + E_y^2} = 10\,N/C$
Hence force $F = QE = 2 \times 10 = 20N$

17) Electric field is perpendicular to the equipotential surface and is zero every where inside the metal.

18) Flux coming out of the cube $\phi_1 = \frac{\lambda . a\sqrt{3}}{\varepsilon_0}$ ...(i)

and from sphere $\phi_2 = \frac{\lambda \cdot 2a}{\varepsilon_0}$ ...(ii)

$\therefore \frac{\phi_1}{\phi_2} = \frac{\sqrt{3}}{2}$

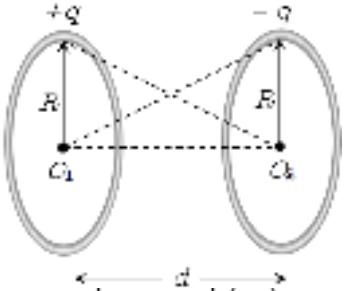
19) Work done by the field

$W = q(-dV) = -e(V_A - V_B)$

$= e(V_B - V_A) = e(V_C - V_A)$ ($\because V_B = V_C$)

$\Rightarrow (V_C - V_A) = \frac{W}{e} = \frac{6.4 \times 10^{-19}}{1.6 \times 10^{-19}} = 4V$

20) Potential at the centre of rings are



$V_{O_1} = \frac{k.q}{R} + \frac{k(-q)}{\sqrt{R^2 + d^2}}$,

$V_{O_2} = \frac{k(-q)}{R} + \frac{kq}{\sqrt{R^2 + d^2}}$

$\Rightarrow V_{O_1} - V_{O_2} = 2kq\left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}}\right]$

$= \frac{q}{2\pi\varepsilon_0}\left[\frac{1}{R} - \frac{1}{\sqrt{R^2 + d^2}}\right]$

21) $\vec{E} = -\frac{\sigma}{2\varepsilon_o}\hat{k} - \frac{2\sigma}{2\varepsilon_o}\hat{k} - \frac{\sigma}{2\varepsilon_o}\hat{k} = -\frac{2\sigma}{\varepsilon_o}\hat{k}$

22) To apply Gauss's theorem it is essential that charge should be placed inside a closed surface. So imagine another similar cylindrical vessel above it as shown in figure (dotted).



23) Potential energy of the system

$$U = k\frac{Qq}{\ell} + \frac{kq^2}{\ell} + \frac{kqQ}{\ell} = 0$$

$$\Rightarrow \frac{kq}{\ell}(Q + q + Q) = 0 \Rightarrow Q = -\frac{q}{2}$$

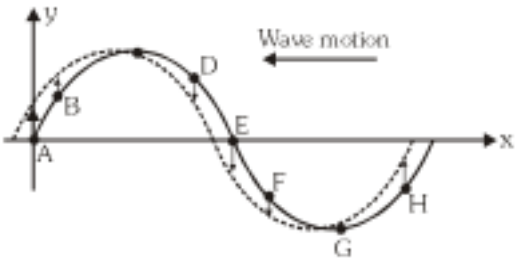
24) K = 0.018

distance between two successive crest is λ

$$K = \frac{2\pi}{\lambda}$$

$$\lambda = \frac{2\pi}{K} = \frac{2\pi}{0.018} = 348.88 \text{ cm} \approx 3.5 \text{ m}$$

25)

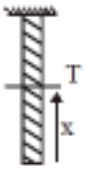


26) process is adiabatic.

27) at x = O, $\frac{dy}{dx} = +ve$ which is present in eq^{n}.

y = a sin(kx–wt) propagating in ⊕ve x-axis

28)



$$V = \sqrt{T/\mu}$$

as tension will keep on increasing

so n will also increase.

29) $\omega = 120\ \pi$, $K = \frac{2\pi}{3}$, $v = \frac{\omega}{K}$

$\Rightarrow v = \frac{120\pi}{2\pi/3} = 180\ m/s$

$v = \sqrt{\frac{T}{\mu}}$ $v^2 = \frac{T}{\mu}$

$T = v^2\mu = (180 \times 180) \times \frac{3 \times 10^{-2}}{1.5} = 648\ N$

30) $y_1 = 4\sin(500\ \pi t)$, $y_2 = 2\sin(506\ \pi t)$

$n_1 = 250$ Hz $n_2 = 253$ Hz

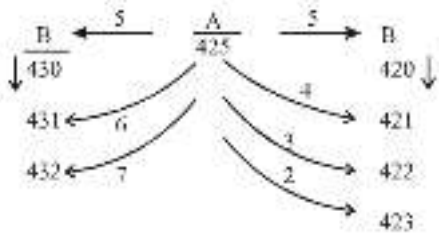
Number of beats = $n_1 - n_2 = 3$ beat/sec.

As $I_1 \propto (16)$ and $I_2 \propto 4$

$\Rightarrow \frac{I_{max}}{I_{min}} = \frac{\left(\sqrt{I_1} + \sqrt{I_2}\right)^2}{\left(\sqrt{I_1} - \sqrt{I_2}\right)^2} \Rightarrow \left(\frac{4+2}{4-2}\right)^2 = \left(\frac{6}{2}\right)^2 = 9$

31) $\frac{3v}{4L_1} = \frac{4v}{2L_2} \Rightarrow \frac{L_1}{L_2} = \frac{3}{8}$

32)



Increasing tension in B, increases its frequency and beats remain '3', so this condition is obtained when B = 420 Hz.

33) Fundamental frequency = difference of two successive resonant frequencies.

So $f_0 = 336 - 252 = 84$ Hz

OR

f_0 = HCF of 252 and 336 = 84 Hz

34) $\nu = \sqrt{\frac{\gamma P}{\rho}} = \sqrt{\frac{\gamma RT}{M_w}}$

$\therefore \nu \propto \sqrt{\frac{\gamma}{M_w}}$

$\frac{\nu_1}{\nu_2} = \sqrt{\frac{7/5}{5/3} \times \frac{4}{28}}$

$= \frac{\sqrt{3}}{5}$

35)

$u_{max} = 2v$
$y_0\ 2\pi n = 2.n.\lambda$
$\lambda = \pi y_0$

36) $f = \frac{v}{2\ell} = 400\ Hz$
$f' = \frac{v}{4\left(\frac{\ell}{2}\right)} = 400\ Hz$

37) The distance of the listener from the second speaker = $\sqrt{(3)^2 + (4)^2} = \sqrt{25}$ = 5 m path difference = (5 – 4.0) m = 1 m
For fully destructive interference 1 m = (2m + l)λ/2
Hence λ = 2/(2m + 1) m
The corresponding frequencies are given by
n = [330 × (2m + 1)]/2 s^{-1}, for m = 0, 1, 2, 3, 4, ...
= 165 (2m + 1) s^{-1}, for m = 0, 1, 2, 3, 4, ...
Therefore the frequencies for which the listener would hear a minimum intensity 165 Hz, 495 Hz, 825 Hz, ...

38) With increase in humidity density of air medium decrease (no change in pressure)
$v = \sqrt{\frac{\gamma P}{\rho}}$, so $v \propto \frac{1}{\sqrt{\rho}}$

39) At constant temperature there is no effect on speed of sound on changing the pressure.

40) Average intensity along the string
$\langle I \rangle = 2\pi^2 f^2 A^2 \rho v$
$\frac{I_1}{I_2} = \left(\frac{f_1 A_1}{f_2 A_2}\right)^2 = \left(\frac{T_2 A_1}{T_1 A_2}\right)^2 = \left(\frac{1}{2} \times \frac{9}{3}\right)^2 = \frac{9}{4}$

41) $y = A\sin^2(kx - \omega t + \phi)$ can be rewritten as
$y = \frac{A}{2} - \frac{A}{2}\cos(2kx - 2\omega t + 2\phi)$
Therefore the amplitude of the wave is A/2 and wavelength of wave is π/k

42) Fundamental frequency $f = \frac{1}{2l}\sqrt{\frac{T}{\mu}}$

$\frac{f_1}{f_2} = \frac{l_2}{l_1}\sqrt{\frac{\mu_2}{\mu_1}}$ (since tension is same)

$= \frac{2L}{L}\sqrt{\frac{\pi r^2\rho}{\pi 4r^2\rho}}$

(since wires are of the same material then r is same)

Hence the desired ratio is 1

43) For the end correction e,

$\Rightarrow e = \frac{l_2 - 3l_1}{2} = \frac{70.2 - 3\times 22.7}{2} = 1.05\text{cm}$

44) Frequency of tunning fork = N

Sonometer wire is either N + 5 or N – 5

$n \propto \frac{1}{\ell}$ $\quad \frac{n_1}{n_2} = \frac{\ell_2}{\ell_1}$

$\frac{N+5}{N-5} = \frac{1.05}{1}$

N = 205 Hz

45) From the equation, k = 0.157

$\frac{2\pi}{\lambda} = 0.157.$

λ = 40 m

But the length of the wire in 4th harmonic,

$\ell = 4 \times \frac{\lambda}{2} = 2\lambda = 80\text{m}$

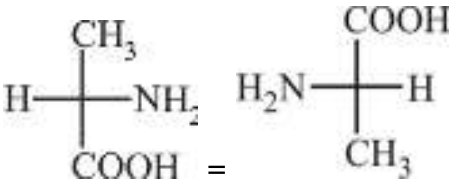
CHEMISTRY

46)

Explain the question: The relative configuration of given compound is.

Concept: This question is based on D-L configuration.

Solution:

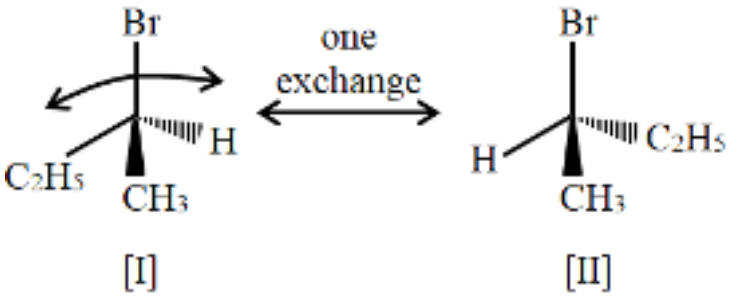


Final answer : 2

47) **Question Explanation :** This question is based on relation between given compounds.

Concept : Enantiomers.

Solution :

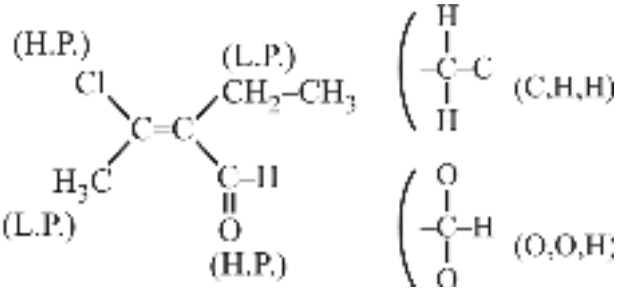


[I] [II]

A. (I) & (II) are having one chiral center and one Can be obtained by one changed one either of them.
B. They are Enantiomers.

Final Answer : Option (1)

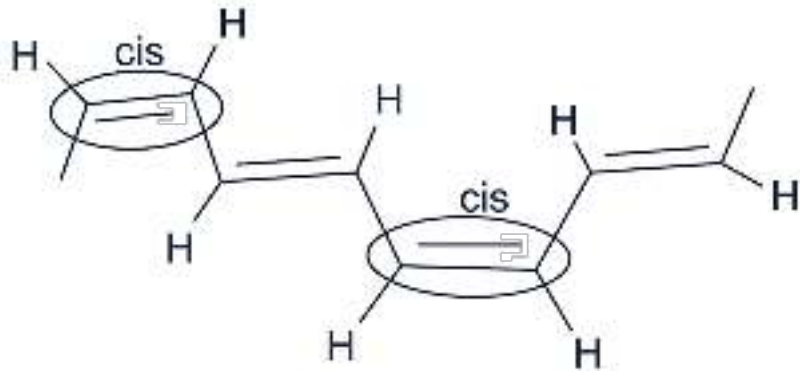
48)



E-Isomer [According to CIP rule]

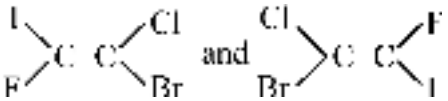
49)

Cis arrangement about double bond has same groups on same side. It is shown below:



Ans 2

50) **Explain the Question :** Identify pair of geometrical isomers.
Concept : This question is based on geometrical isomerism.
Solution : To become pair of GI, valency of one carbon should at same position while position of valencies at other carbon should interchange.



In option (4) pair of GI.
Others are position isomers and identicals.
Final Answer: (4)

51) All are correct

52) $\text{Stability} \propto \frac{1}{\text{Strain}}$

53)

Chiral atom = 0

54)

$2^n = 2^3 \Rightarrow 8$

55)

OA = $2^{n-1} - 2^{P-1}$
Meso = 2^{P-1}

56)

They have opposite configuration

57)

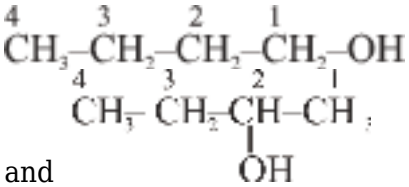
Question Explanation :

Identify the possible number of alcohol with molecular formula C_4H_9OH.

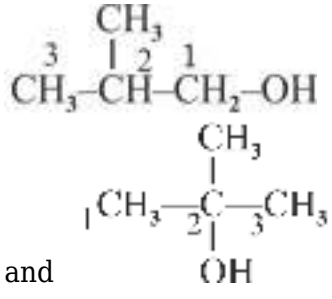
Concept :

Structural isomerism

Solution :



and



and

Final answer : Option(1)

58)

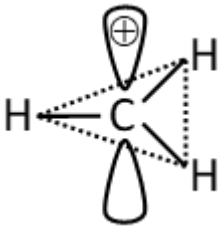
Question Explanation :

Question is asking about True statement about $\overset{\oplus}{C}H_3$

Concept:

Properties of methyl carbocation.

Solution :

• Structure of $\overset{\oplus}{C}H_3$ = 
• It has 3 σ bonds, so carbon is sp^2 hybridised.
• Shape is planar.
• Carbon has 6 valence electrons as it forms three σ bonds.

Final Answer :
Option (3)

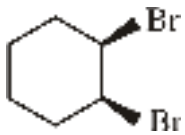
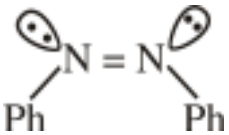
59)

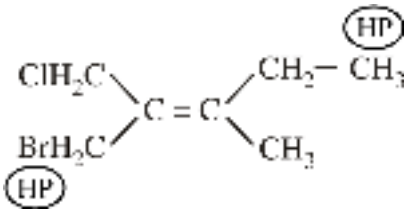
Species having vacant orbital are electrophile

60) $-\overset{\oplus}{N}H_3$ > Cl > –OR > NH_2

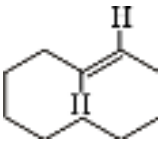
61)

Explain Question: Products set(1) & products set (2) are.
Concept: This question is based on Homolytic and Heterolytic Cleavage
Solution: Homolysis results in the formation of two free radicals.
Heterolysis results in the formation of a cation and an anion. The correct answer is that set 1 contains radicals and set 2 contains.
Final answer: 4

62)  
cis syn



(E)



Trans

63)

NCERT XII haloalkane

64)

Mirror image at D-glucose

65)

According Bredt's rule

66) $CH_2=CH^{\ominus} < CH_2\equiv C^{\ominus}$

67)

$2^n = 2^3 \Rightarrow 8$

68)

For zero order, $t_{1/2} = \frac{a}{2k}$ =and $t_{100\%} = 2 \times t_{1/2}$

$t_{1/2} = \frac{1}{2 \times 0.001} = 500$ sec and T = 1000 sec

69)

$r = K[A]\,[B_2]$

$r = K.K'\sqrt{A_2}\,[B_2]$

$r = [A_2]^{\frac{1}{2}}[B_2]$

70) Slope = $\frac{-E_a}{2.303\,R} = -\frac{1}{2.303} \Rightarrow E_a = R = 2$ calorie

71) $k_1 = \frac{2.303}{t_1} \times \log\left(\frac{100}{100-25}\right)$

$t_1 = \frac{2.303}{k_1} \times$ (log 4 – log 3)

similarly,

$k_2 = \frac{2.303}{t_2} \times \log\left(\frac{100}{100-75}\right)$

$t_2 = \frac{2.303}{k_2} \times$ (log 4)

$\frac{t_1}{t_2} = \frac{k_2}{k_1} \times \frac{(0.6-0.48)}{(0.6)}$ $\left(K \propto \frac{1}{t_{1/2}}\right)$ $= \frac{3}{2} \times \left(\frac{1}{5}\right) = 0.3 : 1$

72)

E_a Ist step highest
So graph (3) is correct

73) $3\,A(g) \rightarrow 2B(g) + 3C(g) + D(g)$

P_0

$P^0 - x \quad \frac{2}{3}x \quad x \quad \frac{x}{3}$

$P_t \Rightarrow P^0 - x + \frac{2}{3}x + x + \frac{x}{3} = P^0 + x$

$x = (P_t - P_0)$

$k = \frac{1}{t}\ell n \frac{P_0}{P_0 - (P_t - P_0)}$

$= \frac{1}{t}\ell n \frac{P_0}{2P_0 - P_t}$

74) $K_1 = \frac{0.693}{20};\ T_1 = 300K$

$K_2 = \frac{0.693}{10};\ T_2 = 320K$

$\log \frac{K_2}{K_1} = \frac{E_a}{2.303 \times R}\left(\frac{T_2 - T_1}{T_1 \times T_2}\right)$

$\log \frac{20}{10} = \frac{E_a}{2.303 \times 2} \times \left(\frac{20}{300 \times 320}\right)$

E_a = 6624 calorie = 6.62 Kcal

75) Rate of reaction decrease by inhibitors.

76) $k = \frac{C_o - C_t}{t}$

$\underbrace{(C_o - C_t)}_{y} = \underbrace{kt}_{mx}$

77) $r_2 = 5.64 \times r_1$
$= 4 \times 1.41 \times r_1$
$= 2^2 \times 2^{1/2} \times r_1 = 2^{5/2} \times r_1$
Also, $r_2 = r_1 \times \mu^{(\Delta T/10)}$ and $\mu = 2$
$\therefore \frac{\Delta T}{10} = \frac{5}{2}; \Delta T = \frac{10 \times 5}{2} = 25$
$T_f = T_i + 25 = 50^\circ C$

78) $k_1 = \frac{2.303}{t} \times \log\left(\frac{100}{100-90}\right)$
$= \frac{2.303}{t}$ (log 10 = 1)
$k_2 = \frac{2.303}{2t} \times \log\left(\frac{100}{100-99}\right)$
$= \frac{2.303}{t}$ (log 100 = 2)
$\therefore \frac{k_1}{k_2} = 1$

79)

$t_{1/2} = \frac{a}{2k}$ (for zero order)

$\log t_{1/2} = \log \frac{a}{2k} = \log a - \log 2k$

80) $k = A \times e^{-E_a/RT}$
$k = A$ (if $E_a = 0$)
$\ln k = \ln A - \frac{E_a}{RT}$; $\ln k$ Vs $\frac{1}{T}$ = linear
(ii) and (v) ⇒ correct

81) Theory based.

82)

$\Delta H = 200 - E_{a_b}$
$-80 = 200 - E_{a_b}$
$E_{a_b} = 280$

83)

E_a is min energy

84) $r_{new} = r_{initial} \times \mu^{(\Delta T/10)}$

$= R_0 \times (2)^{40/10}$
$= 16\ R_0$

85) Since : $t_{1/2} = 15$ min.

- No. of half lives $= \frac{60}{15} = 4$
- Amount of substance left after one hour

$= \frac{A_0}{(2)^n} = \frac{A_0}{(2)^4} = \frac{A_0}{16}$

86)

$X = \frac{1}{1 + 55.5} = \frac{1}{56.5} = 0.017$

87)

$X_{Benzene} = \frac{\frac{7.8}{78}}{\frac{7.8}{78} + \frac{46}{92}}$

$= \frac{0.1}{0.1 + 0.5} = \frac{0.1}{0.6} = \frac{1}{6}$

88) $M = \frac{10xd}{M.M} = \frac{10 \times 98 \times 1.84}{98} = 18.4$ M

89) **Question Explanation:**
The problem asks for the density of a sulfuric acid (H_2SO_4) solution given its molarity (2.55 M) and its mass percentage (20% w/w). We need to relate concentration by volume (molarity) to concentration by mass (percentage) using the density of the solution.

Concept:
The general formula relating Molarity (M), mass percentage (x%), and density (d in g/cm^3):

$$M = \frac{x \times d \times 10}{\text{Molar Mass of Solute}}$$

Solution:

$$d = \frac{M \times \text{Molar Mass}}{x \times 10}$$

$M = 2.55$
Molar Mass = 98
$x = 20$

$$d = \frac{2.55 \times 98}{20 \times 10}$$

$$d = \frac{249.9}{200}$$

$d = 1.25\ g\ cm^{-3}$

Final Answer: Option(1)

90) $K = A \times e^{-Ea/RT}$
if $E_a = 0$, $K = A$ (independent of T)
$\therefore A = 3.2 \times 10^{8}$

BIOLOGY

91)

NCERT XI, Page # 129,130,131

92) NCERT XI Page No. # 128,129

93) NCERT XI Pg. # 126-128

94)

NCERT XI Pg. # 132

95)

NCERT XI Pg. # 148-152

96) NCERT XI Pg.# 149-150

97) NCERT Pg.# 126,127

98)

NCERT XI Pg. # 126,127

99) NCERT XI Pg. # 133

100) NCERT XII Pg. # 133,134

101) NCERT-XI, Pg # 133,134

102)

NCERT-XI ; Page No. # 129-131

103)

NCERT XI Pg. No. # 128,129

104) NCERT XI Pg. # 126-128

105)

NCERT XI Pg. # 126-128

106)

NCERT XI Pg. # 144,148,152

107) NCERT XI Pg. # 144,145,148

108) NCERT (XI) Pg. # 144-148

109)

NCERT XI Pg. # 145-148

110)

NCERT XI Pg. # 145-148

111)

NCERT XI Pg. # 145-148

112)

NCERT XI Pg # 148-152

113) NCERT XI Pg. # 145-148

114)

NCERT XI Pg. # 147-148

115)

NCERT XI Pg. # 145-148

116)

NCERT XI Pg. # 148-152

117)

NCERT XI Pg. # 148-152

118)

NCERT XI Pg. # 144

119) NCERT XI Pg. # 144-145

120) **NCERT XI Pg. # 145-148**

121)

NCERT-XI Pg#4,5,24

122)

NCERT-XI Pg#14,15,16

123)

NCERT -XI, Pg. # 14, 15
Statements A, B, D and E are correct.

124)

NCERT-XI Pg#7

125) **Solution :** Triticum, is a name of genus of wheat plant.
Final Answer : (4)

126)

NCERT XI, Page No. # 6,7
A,B & D are correct

127)

NCERT-XI Pg#8

128)

NCERT-XI Pg#90,91

129)

NCERT-XI Pg#90,91

130)

NCERT-XI Pg#89

131)

NCERT-XI Pg#13

132)

NCERT-XI Pg#13

Heterocyst, Murein, Pili, Chemosynthesis, Nitrogen fixation

133)

NCERT-XI Pg#13,14

134)

NCERT-XI Pg#18

135)

NCERT XI Page No. # 18

136) NCERT XI Page No # 27,28
Red Algae
Polysiphonia, Porphyra, Gelidium
Brown algae
Sargassam, Laminaria, Ectocarpus, Fucus, Dictyota

137)

NCERT-XI Pg#15

138)

NCERT-XI Pg#14,15

139)

NCERT-XI Pg#16

140)

NCERT-XI Pg#17,18

141)

NCERT-XI Pg#19,20

142) NCERT (XIth) Pg. # 27, 28

143) NCERT-XI Pg#27

144)

NCERT-XI Pg#29

145) NCERT-XI Pg#29,30

146) NCERT (XII) Pg. # 32

147) NCERT XI, Pg # 30,32

148)

NCERT-XI Pg#32

149)

NCERT-XI Pg#8

150)

NCERT-XI Pg#8

151) NCERT-XI, Pg # 61,63,64,65

152) NCERT XI Pg. # 68-69

153) NCERT XI Pg. # 65

154) NCERT XI Pg. # 63-64

155) NCERT XI Pg. # 67-68

156) NCERT-XI, Pg # 60

157) NCERT XI Pg. # 75,76

158) NCERT XI Pg. # 64

159) NCERT XI Pg. # 64

160) NCERT XI Pg. # 59

161) NCERT XI-Pg. # 59

162) NCERT XI Pg. # 61

163) NCERT XI Pg. # 60

164)

NCERT-XI, Pg # 59

165) NCERT-XI, Pg # 65,66

166) NCERT XI Pg. # 64

167) NCERT XI Pg. # 74-79

168)

NCERT-XI, Pg # 74

169) NCERT XI Pg. # 78

170) NCERT XI Pg. # 65, 74,75, 76

171) NCERT XI Pg. # 77

172) NCERT XI Pg. # 77

173) NCERT XI Pg. # 74, 77

174) NCERT XI Pg. # 74

175) NCERT XI Pg. # 72,73,74

176) NCERT XI Pg. # 72

177) NCERT XI Pg. # 75

178) NCERT XI Pg. # 64

179) NCERT XI Pg. # 67

180) NCERT XI Pg. # 78,79